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Relative Binding Free Energy Calculations in Drug Discovery: Recent Advances and Practical Considerations

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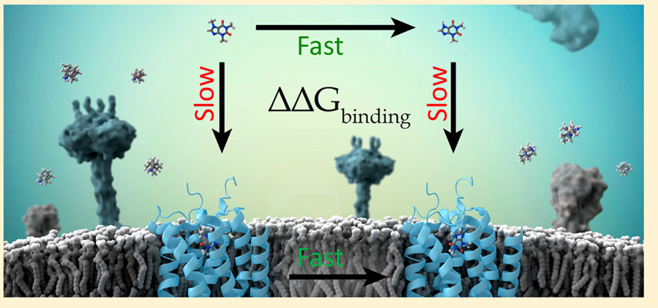
ABSTRACT: Accurate *in silico* prediction of protein−ligand binding affinities has been a primary objective of structure-based drug design for decades due to the putative value it would bring to the drug discovery process. However, computational methods have historically failed to deliver value in real-world drug discovery applications due to a variety of scientific, technical, and practical challenges. Recently, a family of approaches commonly referred to as relative binding free energy (RBFE) calculations, which rely on physics-based molecular simulations and statistical mechanics, have shown promise in reliably generating accurate predictions in the context of drug discovery projects. This advance arises from accumulating developments in the underlying scientific methods (decades of research on force fields and sampling algorithms) coupled with vast increases in computational resources (graphics processing units and cloud infrastructures). Mounting evidence from retrospective validation studies, blind challenge predictions, and prospective applications suggests that RBFE simulations can now predict the affinity differences for congeneric ligands with sufficient accuracy and throughput to deliver considerable value in hit-to-lead and lead optimization efforts. Here, we present an overview of current RBFE implementations, highlighting recent advances and remaining challenges, along with examples that emphasize practical considerations for obtaining reliable RBFE results. We focus specifically on relative binding free energies because the calculations are less computationally intensive than absolute binding free energy (ABFE) calculations and map directly onto the hit-to-lead and lead optimization processes, where the prediction of relative binding energies between a reference molecule and new ideas (virtual molecules) can be used to prioritize molecules for synthesis. We describe the critical aspects of running RBFE calculations, from both theoretical and applied perspectives, using a combination of retrospective literature examples and prospective studies from drug discovery projects. This work is intended to provide a contemporary overview of the scientific, technical, and practical issues associated with running relative binding free energy simulations, with a focus on real-world drug discovery applications. We offer guidelines for improving the accuracy of RBFE simulations, especially for challenging cases, and emphasize unresolved issues that could be improved by further research in the field.



INTRODUCTION

General Overview. Optimization of binding affinity, selectivity, and other off-target interactions is a critical part of hit-to-lead and lead optimization efforts in drug discovery. Relative binding free energy (RBFE) calculations offer an attractive approach to predict protein−ligand binding affinities *in silico* using molecular simulations and statistical mechanics as a way to compute free energy differences between congeneric molecules. From a computational perspective, RBFE simulations are of particular interest due to their rigorous statistical mechanical framework, accurate modeling of biological systems (e.g., protein flexibility, explicit solvent, cofactors, ions, concerted motions, and entropy, to name a few), and direct translation to real-world problems (e.g., hit-to-lead and lead optimization). However, historical challenges have limited the success of free energy simulations in drug discovery—namely, the high computational costs, limited force field accuracy, and technical challenges to setup/run/analyze free energy simulations. The high computational demands have been overcome to a large extent by the recent advances in graphics processing units (GPUs)[1−5] and massive compute resources available on the cloud. In addition, decades of work from academic and industry research laboratories around the world have produced force fields and sampling algorithms that are capable of predicting relative binding free energies at a level of accuracy necessary to be useful in drug discovery.[6−9] Finally, automation tools have addressed many of the time-consuming and error-prone steps associated with running RBFE simulations. These fundamental advances, coupled with workflow automation,[10,11] have enabled free energy simulations to be performed in a rigorous, high-throughput mode that can be readily deployed within structurally enabled drug discovery projects.[12] Furthermore, with the broad availability of open-source (e.g., OpenMM[13] and Gromacs[14,15]), academic (e.g., AMBER,[16] CHARMM,[17] and NAMD[18]), and commercial codes (AceMD[2]

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and Desmond[19]), it is possible to maximally leverage the large number of graphical processing units (GPUs) that are available. The ability to process hundreds of ligands per week running RBFE simulations on a modest GPU cluster opens the opportunity to significantly impact drug discovery projects by exploring much larger portions of chemical space through virtual libraries, of which the best predicted compounds can be prioritized for experimental synthesis and assays. The ability to explore a broader chemical space can lead to molecules with better properties and novel intellectual property (IP), especially for tough to drug targets and in cases where challenging chemistries would not be pursued without compelling evidence to suggest that virtual molecules would indeed have the desired binding properties. While computational chemistry and molecular modeling techniques such as virtual screening, *de novo* design, and evaluation of drug-like properties play many roles in drug discovery,[20,21] here we focus on recent advances and practical considerations for determining protein−ligand binding via RBFE calculations.

The pharmaceutical industry and academic laboratories are increasingly using relative binding free energy (RBFE) simulations in the lead optimization stage of the drug discovery process, as we will show in this perspective, and we expect to see this trend continue as additional successes are reported and computational resources continue to become faster/cheaper.[12,22−24] The growing interest in RBFE simulation methods arises because the relative binding affinity between congeneric ligands can be computed relatively efficiently using a thermodynamic cycle that facilitates a small perturbation of phase space, in contrast to many other quantities of pharmaceutical interest that require much more computational resources (e.g., absolute binding energies, binding kinetics, allosteric effects, biased signaling, etc.). In this work, we do not attempt to give a complete overview of the free energy simulation field; rather, we highlight practical considerations of RBFE simulations that can be used to improve results using current implementations, which we hope will facilitate the successful deployment of RBFE simulations in drug discovery projects. Many of the observations and guidelines presented here have been developed through iterations of internal validation and collaborations, some of which have been already published.[25−30]

Here, we focus exclusively on relative binding free energies (RBFEs) between congeneric ligands—we do not discuss free energy perturbation for absolute binding energies,[31−33] end-point methods (MM-GB/SA, MM-PB/SA,[34] etc.), or other enhanced sampling methods.[35,36] We begin with a short theory overview followed by a discussion of practical aspects of running RBFE simulations, pitfalls, and analysis techniques. We conclude with an overview of recent RBFE applications, including a combination of retrospective validation and real-world prospective uses in drug discovery projects. We examine a variety of factors that can affect the quality of results, including protein preparation, ligand treatment, simulation setup, and the domain of applicability. Additional topics discussed in this work include treatment of kinetically trapped water molecules, force field parametrization, tautomer generation, protein sampling, multiple ligand binding modes, and other RBFE considerations (constructing ligand perturbation maps, mapping perturbed atom, aligning ligands, adding redundancy in the perturbation paths to estimate sampling errors, and inserting intermediate compounds to improve sampling). Additionally, we discuss approaches for monitoring convergence along with proposed strategies to employ if convergence is not achieved. A final point, which is of greater importance to RBFE simulations as compared to approximate and empirical methods, is matching simulation with experimental conditions, since rigorous free energy approaches attempt to simulate the real environment in which the ligands are binding to the protein. Below, we provide a brief RBFE theory overview followed by practical considerations, application examples, and conclusions.

Theory Overview. Ligand binding to a pharmacological target provides the molecular basis for the activity of most pharmaceutical compounds and therefore predicting ligand binding is of critical importance in the discovery of new medicines to treat diseases. Here, we focus on the scenario where one ligand molecule forms a reversible noncovalent stoichiometric complex with the biological unit of a macromolecule at thermodynamic equilibrium with the unbound molecules in solution. The free energy difference between these two end points can be computed by sampling many configurations (microstates), computing the energies of each microstate, and evaluating the partition function Z (from the German "Zustandssumme" or "sum of all states"),[37] which can be used to derive many important properties, such as the binding constant (i.e., binding free energy):

$$RT\ln(K_D) = \Delta G^0_{\mathrm{Bind}} = -RT\ln\left(\frac{Z_{\mathrm{Complex}}}{Z_{\mathrm{Solvent}}}\right);\quad Z_{\mathrm{state}} = \sum_i^{\mathrm{state}} \exp\left(-\frac{E_i}{k_B T}\right) \tag{1}$$

Equation 1 allows the exact calculation for the absolute free energy of binding if the energy for all microstates of each end point (bound and unbound) is known. In practice, computing absolute free energies does not provide an exact match to experiment due to approximations to the energy function and incomplete sampling. First, rather than computing the energetics of the system with quantum mechanics, a classical description via molecular mechanics force fields is introduced to speed up the energy evaluations, at the expense of transferability and accuracy. Even with a quantum mechanical energy model, converging the sampling of all microstates in a protein−ligand complex with explicit water molecules is an extremely computationally intensive task for most biological systems, with 10−100 thousand atoms in a typical system and microseconds to milliseconds of simulated time (billions to trillions of time steps). However, the sampling can be greatly reduced by considering only relative free energy differences between two related molecules, in the cases where the overall binding mode of the ligand is conserved. Fortunately, this approximation is not particularly limiting in the context of lead optimization, where binding free energy differences between two compounds are highly sought. Within the context of relative free energies of binding, a simplification derived from eq 1 can be introduced to reduce the amount of sampling needed, which avoids the need to compute absolute free energies:

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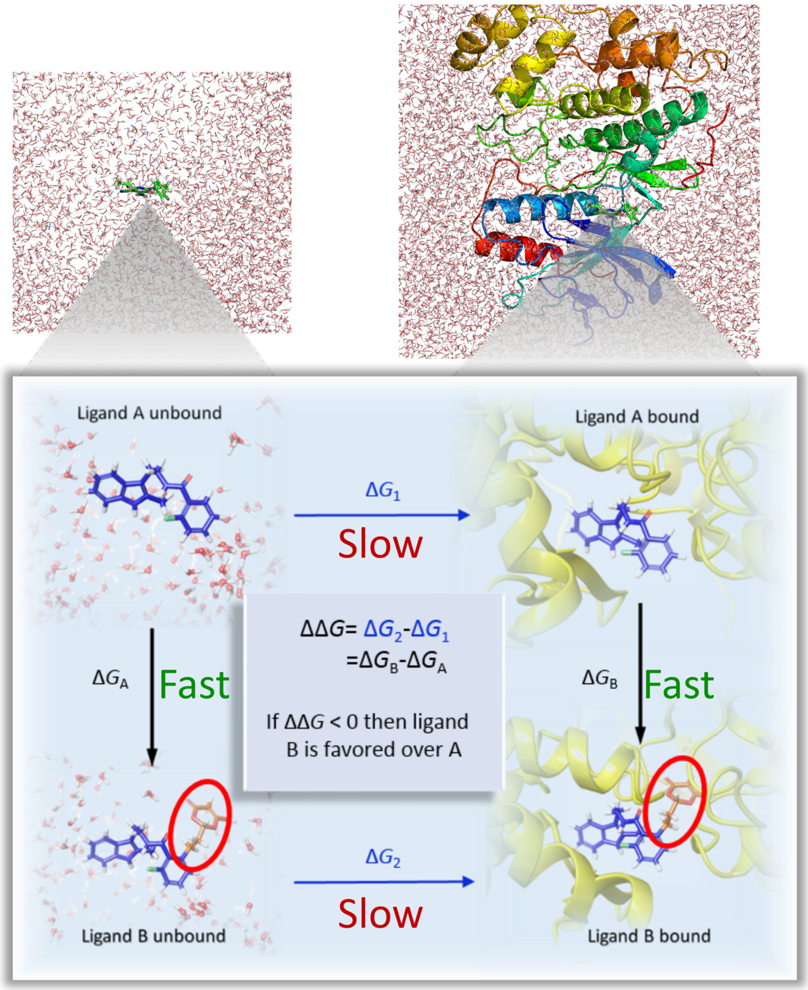

Figure 1. Thermodynamic cycle for relative binding free energy (RBFE) calculations. The relative binding free energy difference between molecules A and B ($\Delta\Delta G$) can be computed via two possible paths. Simulating the direct path (horizontal lines) of $\Delta G_2 - \Delta G_1$ is slow to converge due to the large difference between the end states (water versus protein). The alchemical path (vertical lines) requires much smaller perturbations to the system and therefore tends to converge much faster. Here, ligand A is perturbed to ligand B in the bound state (right vertical line) and the unbound state (left vertical line), the difference of which is identical to the direct path (due to the closed thermodynamic cycle). The top figures are shown to give an indication of the full system size for RBFE simulations, which includes explicit water molecules, while the bottom images are zoomed into the area around the ligands.

$$\Delta\Delta G^0_{\mathrm{A,B}} = \Delta G^0_{\mathrm{B,Bind}} - \Delta G^0_{\mathrm{A,Bind}} = -RT\ln\left(\frac{Z^{\mathrm{B}}_{\mathrm{Complex}} Z^{\mathrm{A}}_{\mathrm{Solvent}}}{Z^{\mathrm{A}}_{\mathrm{Complex}} Z^{\mathrm{B}}_{\mathrm{Solvent}}}\right) = \Delta G^0_{\mathrm{A,BComplex}} - \Delta G^0_{\mathrm{A,BSolvent}} \quad (2)$$

This approach eliminates the error associated with sampling vast portions of phase space that would be needed to compute absolute binding free energies, as it depends only on the free energy change of transforming parts of system A into system B in the bound and unbound states (Figure 1). This so-called "alchemical perturbation" is typically done by connecting the systems via a virtual coordinate, traditionally called lambda (λ), such that a change of λ from zero to one represents a change from system A to system B.

The first application of the free energy calculations to the solvation of organic molecules was performed in 1985 by the Jorgensen group, where perturbation theory was applied to calculate the relative free energies of hydration of methanol and ethane using Monte Carlo simulations.[38] The first ever prospective application in predicting relative free energy of binding of pharmaceutically interesting compounds was performed in 1989 by Merz and Kollman[39] that reported free energy perturbation simulations of a new thermolysin inhibitor, following a retrospective application from Tembre and McCammon.[40]

Free energy calculations can be performed using a variety of schemes such as the thermodynamic integration approach,[41] free energy perturbation using the Zwanzig formula,[42] and the Bennett acceptance ratio (BAR),[43] as well as the multistate BAR (MBAR) variation.[44,45] All these implementations are related to each other,[46,47] but the latter has been suggested to be the most efficient way[48] to conduct free energy calculations and appears to be the most widely used in practice. More details about relative binding free energy calculations can be found in several excellent reviews.[49−68]

PRACTICAL CONSIDERATIONS FOR RBFE SETUP, SIMULATION, AND ANALYSIS

Accuracy and Throughput Considerations. A computational method must be sufficiently fast and accurate to stay

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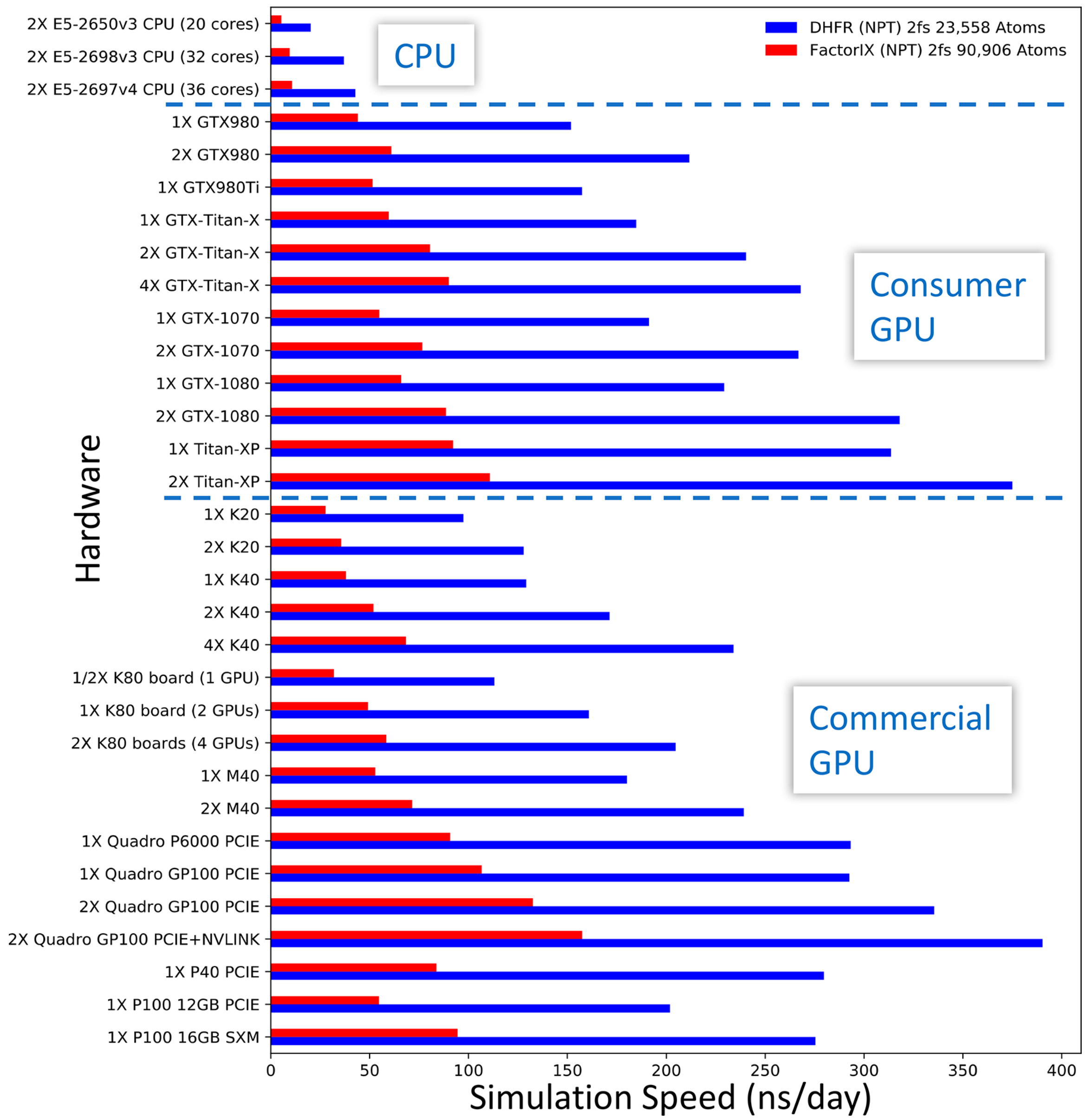

Figure 2. Speed of MD simulations on different hardware for two protein system sizes (23 558 atoms in blue and 90 906 atoms in red), using default simulation parameters with the AMBER16 MD engine.[71]

ahead of synthesis and provide "enrichment" over other approaches to prioritize design ideas. The commoditization of powerful general-purpose GPUs primarily for the video game industry has allowed molecular simulations to be completed in hours (see Figure 2) rather than weeks or months on CPUs, without the need of expensive supercomputers. Performance improvements of molecular simulations using GPUs is especially relevant with algorithms that do not rely on the CPU while running calculations on the GPU, such as for AMBER16.[71] This feature allows for maximum simulation speed and performance because the CPU and PCI-E bus bottlenecks are eliminated, and therefore, the maximum performance of the GPU can be attained. While compute times depend on many factors (GPU architecture, computation protocol, simulation time, MD engine, and the system size), in general a moderately sized 20-GPU cluster can complete dozens of perturbations per day as of the writing of this article (2017) and the throughput will continue to increase with further improvements to GPUs and expansion of cloud computing infrastructures. This throughput is faster and considerably cheaper than experimental approaches (synthesis plus binding assays) and is therefore opportune to integrate into structure-based lead optimization projects in the pharmaceutical industry. However, this throughput, even with a large cluster of GPUs, is far from what would be required in a high throughput virtual screening context to explore millions of molecules. Nonetheless, with the growing availability of GPU resources on the cloud, it is conceivable to run free energy simulations on thousands of compounds per day, which would greatly expand the chemical space that could be explored in a project.

Regarding accuracy, predicted free energies computed using RBFE simulations have been reported with errors around 1 kcal/mol on average in applicable scenarios,[69] which was confirmed in two recent large-scale validation studies.[27,28] A typical ligand modification in a lead optimization project results

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in relatively small changes in binding free energy in the range of 1 kcal/mol.[70] As such, the error in free energy calculations is still too large to eliminate the need for experimental on-target screenings. However, even with errors of this magnitude there is still value in RBFE calculations because they allow one to distinguish between significantly tighter and weaker binders. For example, one can computationally screen virtual analogs of a parent compound until an analog is found with a predicted affinity gain significantly larger than the error in the computation. In this case, there would be a high likelihood that the designed analogs would in fact have an improved binding affinity. Moreover, RBFE simulations have a high success rate in predicting nonbinders, which can be used to eliminate undesired compounds without needing to run experiments. Running massive free energy calculations would vastly expand the explored chemical space within a project and increase the chances of finding promising lead matter. In addition, any improvements in accuracy using better force fields, protocols, or data analysis would make free energy simulation methods even more useful in drug discovery.

System Preparation. *Protein Preparation.* Proper preparation of the target protein to generate a viable starting structure is essential for obtaining reliable RBFE predictions. Some system parameters must be decided upon prior to running free energy simulations because they cannot be calculated by standard simulations and classical force fields, such as tautomer and ionization states of the ligand and protein, which will be discussed below. While sampling during the simulations can resolve a variety of issues (primarily conformational sampling problems), simulation times are typically insufficient to correct major structural errors, such as poorly placed loops, unresolved buried waters, and incorrect ligand binding poses. Even an asparagine or glutamine with an incorrect orientation of the terminal rotatable bond of the side chain (the amide O and N are virtually indistinguishable at standard crystallography resolutions) can destabilize the protein and lead to local unfolding before the correct orientation is sampled, which can adversely affect results, even if the residue is far from the ligand binding site. In some cases, incorrect initial side chain assignments can be remedied during the simulation, but this is not guaranteed; it is always better to start in the correct configuration. In general, free energy calculations are more sensitive to protein preparation as compared to traditional structure-based drug design (SBDD) tools such as docking, where only binding site residues contribute significantly to the results. At the same time, free energy calculations are less sensitive to the exact coordinates of the system, since they can relax during sampling. Therefore, it is essential to prepare the protein such that everything in the system is close to the correct minimum energy basin before starting the calculation. The ability of free energy simulations to work well on inexact structures is exemplified by the robust performance of free energy simulations on homology models, as has been demonstrated recently,[72] although the authors noted that the initial model (including both the protein structure and the ligand binding mode) must be reasonably close to the actual answer in order to get predictive results from free energy simulations.

Other key steps for protein preparation in the context of free energy simulations include adding hydrogen atoms in the correct locations, assigning proper bond orders (especially for the ligands), predicting the location of missing atoms (generally side chains, loops, waters, and ions), optimizing the hydrogen bond network, and minimizing/refining the structure to remove significant clashes present in the crystal structure. The addition of missing atoms in regions of poorly resolved crystallographic density can be particularly challenging and, in some cases, might render a target too challenging for free energy simulations. For example, if there are missing residues around the binding site that cannot be placed with high confidence, then the project might not be amenable to free energy simulations without additional structural biology efforts to generate crystal structures with these atoms resolved. Alternatively, computational predictions can be employed to generate multiple structures, each of which can be run through free energy calculations to determine the structure most likely to be correct by scoring ligands with known activities within each model and determining which one best reproduces the experimental results. While there is no guarantee that the model that produces the best results is indeed the most accurate model, it can be a good indicator. In addition, standard MD simulations can be run on each of the predicted structures to assess which one or more are the most stable and therefore plausible starting points for free energy simulations. The above discussion also holds for different starting crystal structures (i.e., if the choice of a good starting structure is not clear, it can be useful to run retrospective validation studies on multiple starting structures to see which ones produce the best results).

Optimizing the hydrogen bonds (H-bonds) can also be challenging, especially in cases where water molecules are part of an extended H-bond network. In addition to optimizing the orientation of hydroxyl and thiol hydrogen atoms, one must consider 180° flips of Asn, Gln, and His residues where the O/C/N atom locations cannot be distinguished by the electron density. On top of that, His has multiple states (two tautomers and one charged state) that are all close in energy and therefore must all be considered during the preparation process. The ligand might also have multiple protonation states, which are often coupled with the protein states, such that the minimum energy state of either one in the unbound state is different than in the bound state. In such cases, a state penalty must be applied to the free energy predictions for the less stable state in solution to account for the reduced effective concentration of the high-energy species. Finally, H-bond networks can be strongly biased by waters, but water molecules are not always resolved in protein structures. For example, if two carbonyl groups from residues that can flip (e.g., Asn or Gln) are pointing toward each other, it is possible that either there is a bridging water molecule or one of the carbonyl groups must be flipped. In cases where this is suspected, it might be helpful to run multiple MD simulations starting from the different plausible states, as described above, to determine which one is most suitable for starting the free energy calculations. When there is still ambiguity after such simulations, running free energy simulations on multiple states to determine which one produces the closest agreement with known experimental binding energies might be the best strategy. As a general rule of thumb, crystallographic water molecules should be preserved, even those that are far from the binding site, since they may provide stabilization to the system and they improve the equilibration process. More about the treatment of water molecules, especially those that are buried without the ability to exchange with bulk solvent, is provided in the next section.

Another important, albeit often overlooked, consideration is whether there are interactions between the receptor and other biologically relevant protein partners. For example, the

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biological unit might be a multimer, whereas the crystallized structure is a monomer and therefore the biological unit must be built from the monomeric units. In other cases, the crystal structure might have multiple copies of the protein in the asymmetric unit, but the oligomerization state is inconsistent with the biologically relevant state. When there is an inconsistency between the crystallographically determined state and the biologically relevant state, one must generate the relevant state and fix any structural artifacts that might have arisen from the inconsistency. For example, a loop or side chain might adopt different conformations in the crystal structure due to interactions with the crystal partners, which are not present in the biologically relevant state. Such a loop or side chain should have its structure repredicted prior to running free energy simulations, especially if it is near the binding site. Finally, all relevant cofactors should be included, especially when the cofactor is known to play a role in ligand binding. For example, the kinase CDK2 (cyclin-dependent kinase 2) depends on cyclin and therefore cyclin should be included in the system preparation and free energy calculations, even through cyclin is found more than 10 Å away from the kinase active site. Preparation details of this nature are often overlooked when using other computational SBDD methods (e.g., docking) because atoms far from the binding site have no effect on the computations. However, when modeling protein dynamics it is essential to prepare the system as accurately as possible to reflect the true biological state of interest.

Treatment of Kinetically Trapped Waters. Proper treatment of water molecules is critical for accurate and robust relative binding free energy calculations. While a good overall treatment of water molecules is crucial (including the initial protein–ligand equilibration, force field choice, and ligand unbound state simulation), dealing with water molecules in a protein binding site is particularly important and challenging in cases where the buried waters are kinetically trapped and cannot readily exchange with bulk solvent over the time course of the RBFE molecular dynamics simulations, such as for the deeply buried waters in the adenosine receptor A_{2A} GPCR[73,74] or the network of conserved waters in bromodomains like BRD4.[75] Such cases often require special attention and manual efforts to obtain converged and accurate RBFE results.

Replacing a trapped binding site water molecule with a ligand modification can be favorable or unfavorable, depending on the details of the system.[76] Proper treatment of trapped water molecules presents significant challenges for simulations, as has been discussed previously.[77,78] For example, replacing a triazine core with a 5-cyanopyrimidine in Liu et al., displaced a hydrogen-bonded water molecule in the binding site of p38α MAP kinase, leading to a 60-fold enhancement in K_i.[79] On the other hand, there are many examples (most of which are likely not published) where replacing a binding site water molecule results in a loss of potency.[80] Rather than displacing a given binding site water molecule, sometimes it is better to bridge it or even avoid it completely.[80,81] Due to the high degree of water flexibility (translational and rotational) coupled with the fine thermodynamic balance between enthalpy and entropy, it is challenging to *a priori* estimate the energetics of water molecules in a binding site, and even more challenging to estimate the differential effects of ligand binding. For example, Robinson et al. studied a congeneric series of PI3K inhibitors from Sanofi and found that the energetics of water rearrangements in a solvent-exposed region led to changes in affinity that conferred additional selectivity toward their target of interest (PI3K-β) over a closely related kinase isoform (PI3K-δ).[82] These binding sites were very similar and it was hypothesized that the differential affinity arose from an Asp/Asn difference in the proteins over 7 Å from the site of ligand modification, where the charged Asp in PI3K-β was able to preferentially stabilize a water through an extended network in an area where the hydrophobicity increased upon ligand modification. Such energetics are very challenging to accurately estimate without rigorous simulations, but RBFE calculations can fully account for such situations if the relevant water positions are adequately sampled throughout the simulations.

Ideally, buried pockets should have explicit water sampling during the RBFE simulations using approaches that can overcome large energetic barriers to entry, such as Grand Canonical Monte Carlo (GCMC).[83,84] Performing the water sampling within the RBFE simulations using a rigorous approach ensures that the energetics of the water insertion/deletion in the binding site relative to bulk solvent is fully accounted for, which is not the case when waters are trapped and not sampled explicitly during the simulations. However, most current RBFE codes do not have a straightforward way to perform water molecule insertions/deletions during the simulations. As such, using currently available RBFE tools, a good initial configuration of water molecules should be determined prior to RBFE calculations and an estimate of the water insertion/deletion energy should then be added to the RBFE results. Many methods have been proposed to assess the optimal placement and estimated energetics of water molecules, and this continues to be an area of active research. Methods such as 3D-RISM,[85] Aquaalta,[86] Consolv,[87] GIST,[88] GRID,[89] JAWS,[90] SZMAP,[91,92] Waterdock,[93] WaterMap,[94,95] and Water-score[96] have been proposed to predict water positions and energetics, although integrating these methods into RBFE simulations has not been demonstrated. The treatment of kinetically trapped water molecules is an area where improvements would add considerable value to RBFE calculations by improving accuracy overall and expanding the domain of applicability. Approximate methods, such as those listed above, may provide a short-term solution to improve accuracy, but ultimately the water sampling should be fully integrated into the RBFE workflow so that the energetics are rigorously included.

Force Field. In free energy calculations with a rigorous theoretical framework, the accuracy depends only on the degree of sampling and the underlying energy model. Most of the work presented here focuses on sampling, both prior to running free energy simulations (see above) and during the simulations (see below). However, without the right potential energy function, even exhaustive sampling will converge to the wrong answer. As such, having a proper energy model is paramount to the accuracy of relative binding free energy calculations. Using a quantum mechanics (QM) energy model for free energy simulations is ultimately the desired approach but is currently impractical given the available compute resources and throughput needed to make an impact in drug discovery. QM-based molecular dynamics can nowadays be run on a few thousand atoms for a few hundred picoseconds, using massive computational resources,[97,98] which is insufficient to be practically useful in the drug design process where typical systems are tens of thousands to hundreds of thousands of atoms (when considering an appropriate layer of solvating water) and simulation times are on the order of nanoseconds. As such, the potential energy of the atoms in the system is

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typically approximated via a classical molecular mechanics force field where the quantum mechanical forces are approximated by Newtonian physics, which allows for much faster energy evaluations while still representing the underlying physics in a reasonable manner in most cases. Whether the force field representation is good enough in specific cases depends on many factors and may often be unclear. Below, we discuss a few recent advances in force fields, which demonstrate that force fields can be accurate enough for relative binding free energy calculations in drug discovery, although there is still room for further improvements.

One common test of force field performance is the ability to predict absolute solvation free energies of small molecules, which is a good validation method because one can generally sample small molecules in solution sufficiently to converge the free energies and therefore the discrepancies between the calculations and experiment can be attributed to the force field (assuming accurate experiments). In addition, solvation free energies are a surrogate of an important part of the binding process (i.e., a small molecule moving from solution to gas phase loses the water interactions in a similar way as a ligand moving from solvent into a binding site) and therefore of practical interest. Historically, solvation energy errors on the order of 2−3 kcal/mol have been considered to be accurate; however, in order to consistently predict binding energies to within 1 kcal/mol, the error in absolute solvation free energies should be less than that (unless significant cancellation of errors occurs in the binding calculations). In 2010, it was demonstrated that the OPLS force field could achieve an accuracy of 1.3 kcal/mol on a diverse data set of 239 fragment molecules,[99] which was encouraging, but several important functional groups had significantly larger errors (primarily N-containing compounds). More recently, an updated version of the OPLS force field (OPLS2),[100] which contains an improved partial charge treatment and torsional potentials refit to QM, showed a marked improvement in solvation free energies, dropping the root mean square (RMS) error of the data set to 0.9 kcal/mol and significantly improved several of the key outlier functional groups (aromatic amines, bifunctional amines, alkynes, sulfides, ethers, and esters).

In addition to the OPLS family of force fields, there are a number of other well-established force fields that continue to improve ligand parameters. For the most part, the approach taken for force fields is similar: run high-level QM calculations to determine the potential energy surface and then fit the force field parameters to the QM potential. There are many variations in implementation details, but ultimately this approach can be applied to most modern force fields. In some cases, expansion of ligand parameters has been accomplished through automation of transferring existing parameters and generating custom charges, as in the case of GAFF[8] and CGenFF.[9] In other cases, a more explicit approach of generating torsional profiles based on quantum mechanical fits to each molecule of interest has been proposed.[101,102] In addition, limitations in the atom-centered charge model can be overcome with off-center charges, as seen in recent improvements made in CHARMM to the treatment of ligand−protein halogen bonds using the virtual site approach,[103] similar to that employed by Jorgensen et al. in OPLS.[104−106]

The above force fields all share a common functional form with fixed charges. However, it is possible to include a more rigorous QM treatment, as with the X-Pol approach[107] in QMFF.[108] Additionally, Bayesian approaches to force fields apply a prior distribution of parameters to determine posterior estimates of uncertainty in a molecular system, allowing for the determination of both statistical and systematic components of computed results.[109,110] Finally, including explicit polarization is an attractive way to improve force fields and encouraging results have been presented with methods such as AMOEBA.[111,112] Other polarizable force fields have been presented, and reviews on the topic show some advances in the field,[113] but general parametrization for the diversity of ligands in the context of drug discovery lead optimization has been challenging. Indeed, there are many ways in which an approximate force field can be generated to achieve accurate energetics at a fraction of the time of a QM calculation, although all approximations will come at a cost in terms of accuracy. The primary question of interest in drug discovery efforts is whether the gain in computational throughput can justify the loss in accuracy.

The protein force field is also essential in relative binding free energy calculations to accurately model protein motions. Among the force fields most used over the past decade, OPLS has generally performed well for small organic molecules but has lacked high-quality parameters for larger biomolecules (protein, DNA, RNA, and lipids). Recently, the protein parameters in the OPLS force field were improved by refitting to QM using a procedure similar to that performed for small molecules.[6] This refinement overcame limitations associated with older versions of the OPLS force field and put the protein parameters on par with other force fields, such as CHARMM[114] and AMBER,[7] that have been refined over decades of simulations on biological systems and seem to generally perform well for running long time scale MD simulations of proteins. At the same time, force fields like CHARMM and AMBER have continued to improve their small molecule treatment through the addition of more parameters and automated workflows to generate high-quality ligand parameters on the fly.

While an exhaustive review of force fields is beyond the scope of this work, it is important to emphasize that an accurate force field is essential to obtaining reliable free energy predictions. Along these lines, it is helpful to note some of the primary limitations of modern force fields. First, most force fields rely on a fixed charge model, which neglects polarization. Even if polarization can somehow be incorporated into the charge fitting procedure (either explicitly or implicitly), the functional form limits the polarization from changes in the environment. Ideally, a force field would adapt, as nature does, to the changing environment, be it in solvent, protein, lipid, or any other environment. Along these lines, the fundamental form of most force fields rely on approximations that proved to be fast and reasonably accurate but has not been updated in decades; e.g., it inadequately treats close contacts that cannot be explained with a Lennard-Jones potential or cross terms between the parameters. Next, the atom-centered charges neglect directionality that is needed to represent charge distributions for lone pairs,[6,115] halogen bonds,[106,116,117] and other sigma hole interactions,[118,119] although some improvements have been made through the introduction of off-centered changes, as discussed above. Another significant limitation of most modern force fields is the reliance on atom types, which have been constructed to simplify the process of assigning bonded and nonbonded parameters to atoms that should have similar properties, but this generalization creates a lack of precision on the specific parameters applied to each atom. It

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should also be noted that even if a framework exists to address the above problems, there are still challenges associated with automating such a workflow to be robust in the context of industrial applications where hundreds of ligands need to be processed in a single iteration of lead optimization. Finally, building a "good enough" force field for RBFE calculations in drug discovery lead optimization will likely need to be done iteratively using accurate binding free energy data, which is still sparse. Updating ligand parameters alone is likely to be insufficient, since the binding properties of a ligand depend on interactions with water in the unbound state and protein in the bound state, and attaining the precise balance between all of these terms is extremely challenging and time-consuming.

Sampling Considerations. *Treatment of Protein Flexibility.* A major advantage of MD simulations is the ability to capture flexibility, especially concerted motions, among ligands, proteins, solvent, and any other molecules in the system. This enables MD to accurately treat important factors underlying biomolecular recognition (e.g., induced-fit, solvent rearrangement, and entropy). While it is also possible to simulate large-scale and slow conformational changes, such as protein folding,[120] secondary structure changes,[121] and ligand binding kinetics,[122] in practice there is a time scale limitation that restricts the types of events that MD can treat in drug discovery applications using commodity hardware. The time scales of protein dynamics[123,124] suggest that a multinanosecond MD protocol can adequately sample local conformational deformations, water equilibration (when not buried), many protein side-chain motions, and at least some higher-energy transitions (loop motions and high-barrier rotamer transitions), but protein domain rearrangements, folding, and binding kinetics are currently beyond the reach of MD when quick turnaround is needed.

(A) Backbone Flexibility. Accounting for backbone flexibility may not be required to predict the relative free energies of binding between congeneric ligands, as demonstrated by the accurate results obtained using molecular dynamics free energy simulations coupled with Monte Carlo (MC), where a fixed protein backbone model is generally employed.[67,125] However, one cannot know in advance whether protein backbone flexibility is needed to obtain accurate results for a given system, and in reality at least some backbone motion is required to fully capture all entropic components of binding, even when the average backbone coordinates remain largely unchanged between ligands. For example, in the protein complex Arp2/3 it was shown that the flexibility of the system plays a crucial role in the interplay between ligand and protein, as it influences ligand location within the binding site. Arp2/3 has two different allosteric sites: one at the interface between the Arp2 and Arp3 subunits (pocket 1) and one within the Arp3 subunit (pocket 2).[126] FEP/MC calculations for congeneric ligands within pocket 1 showed good correlation between experimental and calculated results.[30] However, FEP/MC calculations in pocket 2 failed to obtain any significant correlation with experimental data. While several solutions were sought (increasing the sampling steps, inserting more intermediate lambda simulations, starting the calculation from different configurations), good results were only obtained after enabling full protein flexibility via MD (unpublished results).

More evidence of the relevance of backbone flexibility can be seen in the context of membrane proteins. In a previous free energy simulation study of the relative binding affinity of aminoadamantane inhibitors of the Influenza A M2TM pore, removing the membrane and using backbone constraints on the protein in aqueous solution showed a dramatic deterioration of the results. Using a fixed backbone provided poor correlation between the calculated and the experimental binding affinities (R^2 = 0.20), while performing the same calculations using a fully flexible system, with the A/M2TM protein embedded in a DPPC lipid bilayer, improved the correlation to R^2 = 0.85.[127] The low correlation observed here when freezing the protein backbone can be attributed to a dependency of the free energies of binding on the motions of the protein structure, as also discussed elsewhere.[128] The fact that the results degrade when the system is not set up correctly (i.e., neglecting the membrane or freezing parts of the protein) further demonstrates that it is important to build a realistic atomic scale model to get robust results.

A recent validation study of free energy calculations in fragment optimization (see the supporting material of ref 28) contains examples of protein backbone conformational changes influencing the modeling results. For a series of Janus Kinase 2 (JAK2) inhibitors, two protein X-ray crystal structures were available, cocrystallized with ligands of different size, resulting in a deformation of a small part of the binding pocket. Both receptor structures were independently used as inputs and produced very similar and accurate results (R^2-values of 0.8 between calculations and 0.6 comparing either calculation to experiment). However, placing large ligands into the binding site cocrystallized with a smaller ligand required constraints in the docking calculations to ensure the correct binding mode. This case showed that while the protocol is quite robust with respect to different input structures, starting with an accurate ligand binding mode was critical to obtaining accurate predictions. Improvements to docking algorithms, especially accounting for protein flexibility and explicit waters, would alleviate the need for manually biasing poses, as is routinely done today. A similar case involving fragment inhibitors of HSP90 is reported in the same study. Again, two X-ray crystal structures with ligands of different size were available and showed evidence of a helix partially unwinding in the context of certain ligand substituents. Here, only one of the receptor structures (that of a larger ligand) could be used for modeling, since all members of the ligand series could only be reliably placed in that receptor binding pocket (many ligands were too large to fit into the smaller binding sites). A third example from that same work involves fragments binding to p38α MAP kinase, where again multiple receptor cocrystal structures were available and showed major protein backbone conformational motions in parts of the binding pocket. In this case only a subset of ligands of comparable size could be used in calculations, since no single receptor conformation could be used to accurately dock all ligands in the series. These examples show that even moderate protein backbone conformational changes can introduce additional hurdles and uncertainties into predictions. Until a more robust and routine way to treat protein backbone flexibility during docking is developed, a case-by-case treatment of such systems seems to be necessary.

(B) Side-Chain Flexibility. A common kind of protein motion observed upon ligand binding is the rearrangement of one or more side chains,[129−131] which sometimes can happen far from the binding site. Such changes are important to model in RBFE simulations and a molecular dynamics approach offers many direct and indirect ways to account for such motions. In addition, a wealth of information can be extracted from MD simulations, which can be used to understand the protein side

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chain dihedral dynamics,[132] although such approaches are often underutilized in applied simulations. In some cases, a single side chain can significantly influence results, as demonstrated by Buijnsters et al.,[133] where they ran free energy simulations on a series of pyrazine PDE2 inhibitors and found good agreement between calculations and experiment for all but one of the compounds. The single underpredicted compound was the only one lacking a large side chain at a specific ligand position. Experimental X-ray structures showed that this compound induced a conformational change in a key binding site leucine residue that filled the vacant pocket left from the smaller ligand substituents. It is known that for this system, different ligands can induce either an open and closed conformation of parts of the binding pocket, based on the rotamer of Leu770.[133] However, the conformational transition between these states was only observed consistently after 30 ns of MD simulation, which is significantly longer than RBFE simulations are routinely run at the time of this Perspective article. This is an example of a case where protein starting structure bias can be overcome with longer simulations, although without *a priori* knowledge of the two states it might not have been known that additional simulation time was needed. A detailed study of single side-chain rotamer influences on binding free energy predictions for the L99A mutant of T4 lysozyme, a deeply studied test system, can also be found in work by Jiang and Roux.[134] Figure 3 shows an example of a relatively small change in the position of a single side chain can have a significant effect on the binding mode of a ligand.

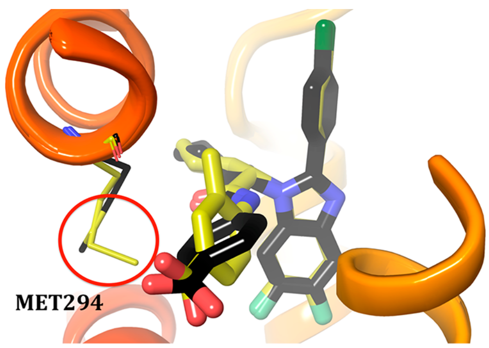

Figure 3. Example of a protein side chain in farnesoid X receptor (FXR) that can adopt multiple conformations based on the bound ligand. Met294 (see red circle) adopts an open or closed conformation, depending on whether or not the inhibitor has an ortho substituent. Carbon colors match between the ligand and corresponding protein (yellow ligand with yellow FXR side chain and black ligand with black FXR side chain).

Another example of the importance of side chain sampling in RBFE simulations can be seen in work by Wang et al.,[27] where the authors showed how coupling free energy simulations with an enhanced sampling scheme (replica exchange with solute tempering; REST)[135] produced better convergence of the predicted binding free energy for a set of ligands bound to CDK2. By comparing with a standard free energy simulation approach, the trajectories showed that REST increased conformational sampling of Lys88 and Lys89, which are adjacent to the ligand modifications of interest, thus providing a better-sampled ensemble and therefore more converged free energies. In this case, it was proposed that the initial ligand pose was making incorrect hydrogen bonds with the protein, and breaking those hydrogen bonds to make the correct ones required significantly longer simulation times or enhanced sampling. Cases like this illustrate the importance of careful side chain treatment, even when there are no obvious clashes that would force significant side chain movements.

Equilibration. Another important consideration that can affect free energy calculation results is the equilibration protocol employed after preparation of the protein and ligand systems and before the free energy simulations. Systems that have not been sufficiently equilibrated in relative binding free energy calculations can lead to large convergence errors due to incomplete sampling of the relevant microstates (and perhaps even missing some relevant microstates), propagating error from the energies collected before the system finds equilibrium. Systems that are not sufficiently equilibrated can also diverge in the phase space overlap of their intermediate states along their perturbation path, leading to suboptimal results. Therefore, it is important to sufficiently equilibrate a starting ligand both in solution and bound to a protein of interest to provide the best starting point for the free energy calculations.

Standard equilibration protocols have been developed, but each protein−ligand system must be inspected to determine if the system has reached an acceptable equilibrium starting conformation. This step is extremely important for accurate relative free energy results due to the dependence on this starting conformation for each lambda run that is performed, and ample time should be spent determining if the equilibration protocol employed was sufficient, or if more sampling should be performed before starting the production runs. For example, relative free energy calculations with AMBER Thermodynamic Integration (TI) on the c-Jun N-terminal kinase 1 (JNK1) significantly improved when increasing the equilibration protocol from 50 ps to 1.4 ns (mean unsigned error reduced from 1.37 to 0.90 kcal/mol; unpublished data). This system involved 31 perturbations across 21 unique ligands. The initial 50 ps equilibration protocol involved 200 steps of steepest descent minimization, 10 000 steps of the NVT ensemble heating from 5 to 300 K, and 10 000 steps of the NPT ensemble in four iterations where restraints were slowly released. The step size was 1 fs. This equilibration was run for the ligand only in solvent starting with restraints on the ligand, and the protein−ligand complex in solvent starting with restraints on the ligand and backbone. The new equilibration protocol that significantly improved the accuracy of relative free energy estimates followed the same equilibration protocol except the NVT ensemble was run for 600 000 steps and the same four iterations of the NPT ensemble were run for 200 000 steps. Note that intermediate equilibration protocols resulted in errors between 0.90 and 1.37 kcal/mol, suggesting that the length of the equilibration protocol was correlated with reduced error. This example demonstrates that with all other variables held constant, a longer equilibration protocol can improve accuracy in the relative free energy predictions. Determining how much equilibration is enough is a challenge that has not been adequately addressed in the literature and is likely somewhat system dependent, with variables such as protein flexibility, quality of the starting crystal structure, and system setup details being some of the many factors involved.

Treatment of Ligands. *Choosing a Reference Ligand.* Alignment of virtual molecules to a reference pose prior to relative binding free energy simulations is crucial for obtaining accurate predictions. All analogs should be aligned to a ligand with a crystal structure, when available. When multiple crystal structures are available, the one containing a ligand with the

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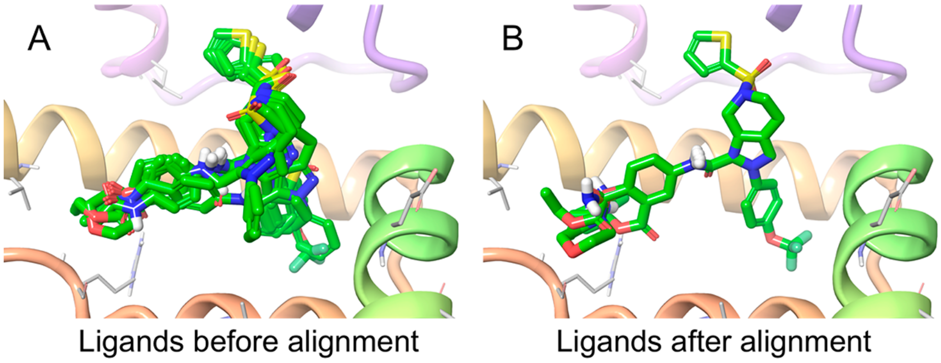

Figure 4. Example of computed poses for a congeneric series of ligands. (A) Docking tends to produce poses that fit well into the receptor structure but do not have a high overlap of the common core. (B) Poses after alignment to a reference crystal structure ligand using maximum common substructure (MCS) superposition produce a perfect overlap of the common core atoms but may contain clashes with the protein. Bad contacts for the new ligand functional groups can be alleviated by an energy minimization or short MD equilibration, which should maintain the ligand and receptor structure in the final pose. A combination of MCS alignment for the core and sampling of the modified substituents may offer the best balance for initial pose placement. Example from the target FXR (PDB ID: 5Q0I).

highest similarity to the virtual molecules should be used.[136] After alignment, some ligands are likely to have bad contacts with the receptor, hence conformational sampling and/or energy minimization should be applied to check whether these clashes are severe or can be alleviated by maintaining the ligand and receptor structures in the final pose (see Figure 4).[137] A careful equilibration protocol should then be applied to get the ligand, protein, and waters into a state without significant problems. The equilibration protocol will depend to some extent on the system of interest and the changes being made to the ligand, but standard equilibration protocols have been validated by many groups.[78,138−140]

Pose Placement. Starting with reasonable binding poses for ligands in the series is a critical step in generating accurate binding free energies. Inaccurate binding modes in the best-case scenario will lead to slower convergence; in the worst case, it will lead to incorrect results. Indeed, sampling of high-energy barriers (ligand torsions, different subpockets, different binding modes, etc.) can be computationally demanding using molecular dynamics without any sort of enhanced sampling. In many cases, pose placement for RBFE simulations involves a core-constrained docking calculation with the core of the modified compounds constrained to the core of the lead compound, which ideally would come from an experimentally determined structure (e.g., from X-ray crystallography). In cases where an experimental structure is not available, which can be common in real-world drug discovery projects, it may be necessary to include protein flexibility in the docking procedure. While a variety of approaches have been proposed to incorporate protein flexibility, including all-atom explicit solvent MD simulations,[122,141] such approaches are still time-consuming and have not been broadly validated to consistently produce better results than simpler/faster methods. Rather than explicitly including receptor flexibility during the docking, it may be possible to use an ensemble of protein conformations, as has been proposed by others,[142−145] although significant challenges still exist for this approach.[146] Specialized docking algorithms to prepare structures for RBFE simulations are not common in the field but would be of high value. In many ways, docking for RBFE simulations is significantly easier than traditional docking of unrelated molecules, since in most cases one can reasonably assume that the core of the molecules within the congeneric series assumes the same binding mode. As such, sampling can be focused on the variable parts of the ligand and the closest receptor residues.

The requirement for a conserved binding mode is a limitation for RBFE calculations, although in most lead optimization efforts the ligand-binding mode is conserved throughout the ligand in the series. For cases where results are inaccurate, convergence cannot be obtained, and/or an alternate binding mode is suspected, several approaches can be pursued. First, enhanced sampling methods could be applied to the degrees of freedom considered to be important for sampling the correct conformational ensemble. In many cases one can infer the groups of the ligand or protein that need enhanced sampling, which typically involve those directly around the region of modifications within the chemical series. In cases where enhanced sampling methods fail to generate converged results or a reasonable initial binding mode cannot be obtained, a new cocrystal structure should be pursued. In cases where that is not possible (or prohibitively slow for project timelines) longer MD simulations can be run to explore alternative binding modes. One can consider such simulations to be extensions to the RBFE equilibration protocol. Such MD simulations could be run on the new molecules in a traditional way (i.e., standard MD) or a hybrid topology could be used for the MD simulations, which has the drawback of potentially moving the protein−ligand structure away from the initial conformation that is correct for the lead compound but ensures that the cores of the molecules overlap, which is a requirement for most RBFE implementations. In general, the limitation of a well-conserved binding mode in a series is necessary to ensure that sufficient sampling can be performed in the relevant region of phase space and that sufficient overlap between adjacent lambda windows can be obtained.

Due to the above-mentioned limitations, great care should be taken to ensure that functional groups within the series of interest should be properly placed, especially for cases with high barriers of rotation (either due to internal barriers or from interactions with the protein).[147] While automated docking tools can be reliable in lead optimization scenarios where a crystal structure with a ligand from the series is available, especially considering that the core is constrained in the docking, manual inspection of the poses is still recommended prior to running free energy simulations due to the high computational costs of RBFE simulations.

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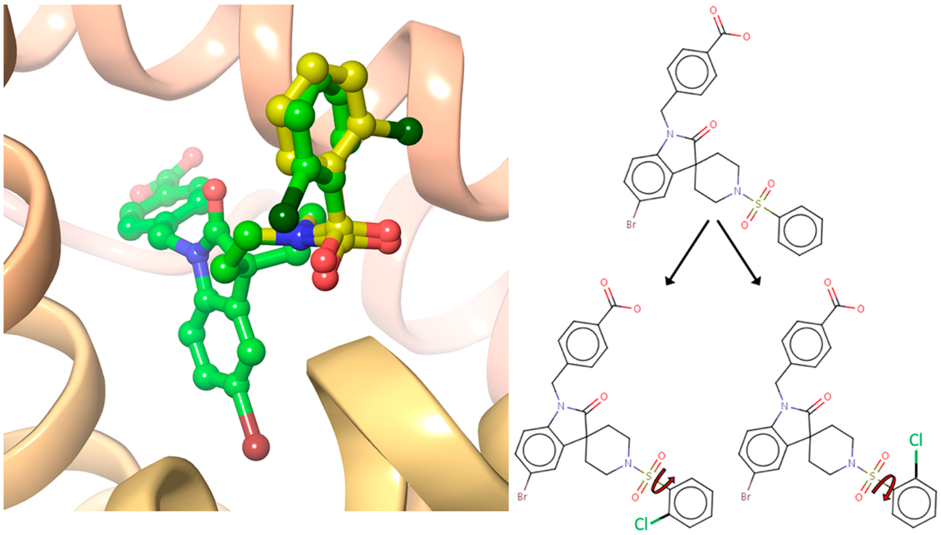

Figure 5. Example of multiple ligand occupancy in a crystal structure. (A) Sulfonamide ligand bound to FXR displaying multiple occupancy for the ortho-chloro substituent (PDB ID: 5Q0W). (B) Multiple RBFE calculations can be used to distinguish between poses related by a 180° bond rotation (i.e., when an ortho- or meta-substituted phenyl group has two possible positions) by performing both perturbations from the unsubstituted compound. Red curved arrows denote the hindered rotatable bond; the size of the chloro group has been increased for clarity.

Accurately predicting binding modes for congeneric ligands in preparation for RBFE simulations is an area where improved methods could offer great value within the drug discovery environment. Indeed, manually inspecting, fixing, and refining hundreds or thousands of ligands on a weekly basis is a laborious and inefficient process, but at the time of this publication it is still highly recommended until more robust methods for pose prediction of congeneric ligands are available. Such methods will likely require some degree of protein flexibility in addition to ligand sampling in order to obtain reasonable starting points for RBFE simulations. Utilizing information about known ligand binding modes has been shown to significantly improve docking results,[148,149] which suggests that a docking protocol focused specifically on congeneric ligands, possibly including more flexibility in the protein around the ligand perturbations and less sampling of the ligands themselves, would be a valuable avenue for researchers to pursue.

Multiple Binding Poses. A good starting ligand binding mode is an essential component of performing accurate RBFE calculations, as the initial structure plays an important role in adequately sampling the relevant regions of phase space and therefore critical to obtaining high-quality RBFE results. In some cases several conformations of the ligand are important for binding,[60] some of which may be observed within the free energy calculations sampling time, but others may not be accessible due to their separation by high-energy barriers. If different conformations rapidly interconvert during the simulation, the same $\Delta\Delta G$ is expected for all of the input ligand conformations; however, if high energy barriers separate different putative ligand binding poses, then alternate strategies should be taken. For example, one may perform independent RBFE calculations considering each binding mode in different simulations. The ligand binding modes that have significantly higher relative binding free energy can be discounted from the results, assuming that the free energy calculations are converged and there is overlap in the phase space between the ligands in the RBFE simulations. The pose(s) with the most favorable free energy of binding implies the dominant binding mode and should be kept for further analysis. For multiple binding poses that exhibit comparable free energy, one can correct the binding free energy by taking into account the contribution from each binding mode into the total binding free energy prediction using Boltzmann weighting, as shown by Kaus et al.[150] Note that in addition to confirming adequate phase space overlap between the ligands of interest (as necessary for all RBFE simulations), it is essential in such cases to ensure that the ligand samples both conformations in the unbound state in order for the energies to be comparable. Complete sampling of the relevant conformations in the unbound state is generally a requirement for accurate RBFE simulations, although in the cases of multiple binding modes it is more important, since a single binding mode increases the chance for cancellation of errors if a low-energy unbound conformation is missed across all RBFE simulations within the congeneric series.

Another way to predict the relative binding free energy difference between the poses and assess the pose with the lowest binding free energy, is to use free energy simulations to transform the different ligand binding poses into each other.[151] For example, a possible scenario is to distinguish between two poses of an ortho or meta substituted phenyl ring after a ring rotation of 180° (Figure 5A). When these two ligand binding poses are identical other than a symmetry related transformation, a free energy transformation can be conducted introducing the unsubstituted phenyl as an intermediate and therefore getting the energetic difference between the two symmetry-related states (Figure 5B). In addition to predicting which of the binding modes is preferred, the overall binding of the substituted compound can be correctly accounted for by introducing a symmetry correction term, which can be up to 0.6 kcal/mol ($kT \ln(2)$) when the two states are isoenergetic, and greater if there are higher degrees of symmetry.[152]

Tautomer/Ionization States. Tautomerism occurs when two or more structures can interconvert by an intramolecular movement of a hydrogen from one atom in the molecule to another. Different tautomers can have significant differences in their contribution to binding due to changes in interactions (H-bond donors becoming acceptors, and vice versa) and energetic

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costs associated with generating the different tautomers. Indeed, it has been shown that ∼25% of a database contains molecules with multiple tautomers that should be considered in the context of protein−ligand binding.[153] The accurate prediction of tautomeric states (and their associated energies) is a challenge for many aspects of structure-based drug design and plays an especially critical role in RBFE calculations due to their sensitivity to modeling the correct underlying physics/chemistry of the binding process of interest. In general, tautomers should be treated during the ligand preparation step because sampling of such states is not part of standard MD simulations. Whenever there are multiple possible tautomer states for a ligand of interest, it is worth considering each state as a distinct ligand and introducing a perturbation between the forms. When the tautomers have different energies in solution, an energetic offset should be added to the computed binding free energies to account for the different concentrations of the two (or more) states in solution. Similar to the above discussion about multiple binding modes, all possible tautomeric forms should be considered and Boltzmann weighted in the final energy prediction, since experimentally all tautomer and ionization microstates of a single compound are present and contribute to its overall binding energy. However, when one state is significantly more favorable than the others, then that state will dominate the total binding free energy and thus a Boltzmann sum will not be significantly different from the energy of the most favorable state.

It is worth mentioning that treating different protonation states (ionization states), and therefore different net charges, presents a significant challenge for RBFE calculations and will be discussed in the next paragraph. Thus, while it is theoretically correct to include different energetically accessible ionization states, from a practical perspective it may add more signal than noise. On the other hand, energetically accessible tautomeric states should always be included and treated as separate ligands when running RBFE simulations, since the "correct" state is only known after the results of the RBFE simulations are combined with the tautomeric energies. The use of methods like constant-pH molecular dynamics has been proposed for including tautomerization in free energy calculations,[154,155] which would remove the need for pregenerating the relevant tautomers, although validation of this approach in the context of RBFE simulations has been limited.

Charged Mutations. In general, the total charge of the ligands within a set of perturbations should remain conserved when performing mutations to ensure the most reliable results. While it has been reported that changing the total system charge can yield accurate results,[156] other studies have suggested otherwise,[127] and limited examples of charge changes exist in the literature. It has been shown that turning on or off a charge during a free energy calculation is nontrivial when using particle-mesh Ewald (PME) for the treatment of long-range electrostatics.[157,158] The problem arises mainly from the fact that a charge change in PME is handled by introducing a uniform neutralizing background charge to enforce neutrality,[159] which may not be appropriate for comparing the binding energy of molecules with different charges. This means the calculated free energy is the sum of turning off/on the charge while simultaneously turning on/off a uniform neutralizing background charge, which can introduce an error in the overall difference in the binding free energy (likely due to not modeling the effects of explicit counterions that can be localized around the charged group). Additionally, charged groups produce much larger contributions to the energy, which will not cancel in the case where the charge is not conserved across the ligands. Even errors in the solvation step (perturbing ligand A to B in solution) could result in many kilocalories per mole of error, which will be compounded with the differences in the bound state energy.

Additionally, the time needed to rearrange the surrounding solvent network might require considerably longer simulation times in the case where the net charge changes between ligands, and overlap between adjacent lambda windows will likely be lower than for charge-preserving changes. A more appropriate treatment that would better reflect the underlying biophysics of the system is to include explicit neutralizing ions that would equilibrate to the correct locations for each ligand in the perturbation. However, equilibrating ions in this way can be very time-consuming and it has not been demonstrated that this can be done effectively in the context of perturbations to ligands that involve changes in net charge. In addition, as mentioned above, it is not clear if sufficient overlap between adjacent lambda windows would exist in the case where a neutralizing ion is needed for one of the ligands and not for the other. An alternative to treating the ions explicitly would be to include some form of correction based on a continuum electrostatics model. For example, Poisson−Boltzmann (PB) calculations have been investigated by Rod and Brooks to remove periodic boundary condition artifacts for the protonation of dihydrofolate in DHFR,[160] although more general schemes are also under investigation.[161,162] Nonetheless, there is limited validation data on any methods for treating charge changes in RBFE calculations, and as such, it is recommended at the time of this publication that such perturbations be avoided. Instead, when charge changes are thought to be important to improve a ligand, such molecules should be made experimentally and if they are found to be better, then they can form the basis for a new congeneric series of ligands with the same formal charge that can be compared with RBFE simulations.

Atom Mapping Between Ligands. Atom mapping in single-topology relative experiments, if not done properly, can be a source of large errors due to the need for sufficient overlap in phase space between neighboring lambdas along a perturbation path and the need to maximize similarity between states by removing/decoupling as few atoms as possible. The most common approach to atom mapping includes using maximum common substructure (MCS) to align atomic graphs regardless of atom type and bond valences, thereby maximizing their topological overlap. In the case where two molecules do not differ in the number or connectivity of atoms regardless of atom or bond type, the mapping task is relatively simple as there is no need for atoms to disappear or appear along the perturbation path and therefore convergence errors are often minimized (e.g., in the case of changing a hydrogen atom to different halogens).

When there is a difference in topology of the two molecules of interest (i.e., the MCS differs between the molecules), atoms not present in the atomic graph of one molecule are introduced as dummy atoms to match the topology of the other molecule. These dummy atoms are defined as place holders that initially have no charge or Lennard-Jones interactions with the remainder of the system, but retain their bonded interactions so that they are in reasonable positions when their potential is turned on. Throughout each lambda of the alchemical path the

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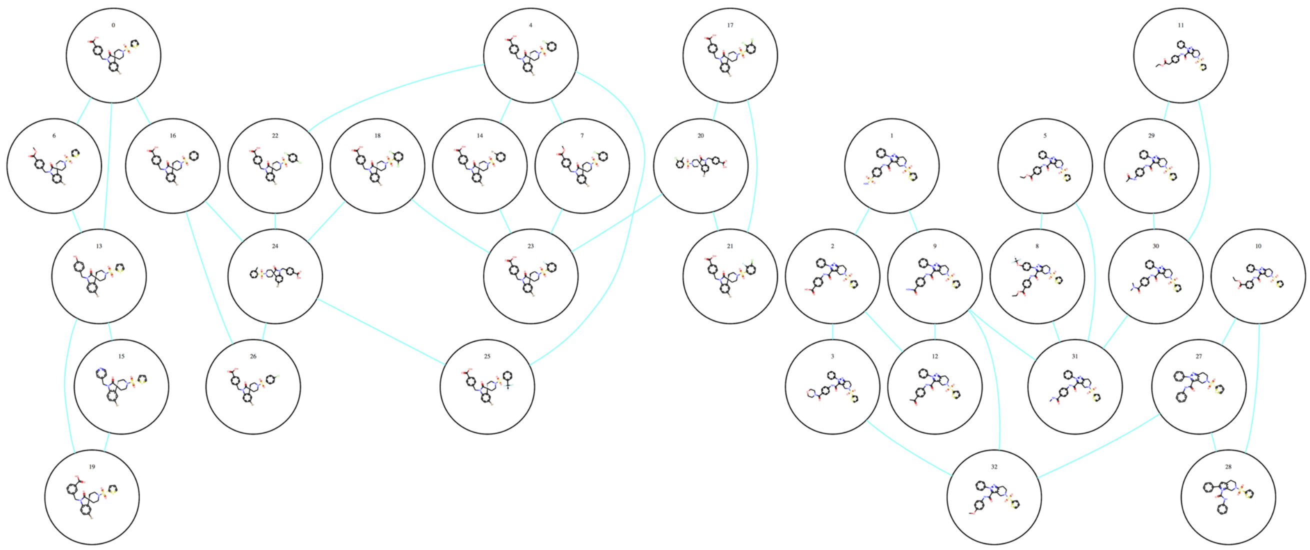

Figure 6. Example of an RBFE perturbation map for FXR ligands using the LOMAP program.[11] This case shows multiple connections between each ligand, which adds redundancy and allows for sampling error estimates as well as more accurate predictions. Note that not all perturbations need be directly connected to the reference ligand, although errors accumulate as more edges are traversed to get from the reference ligand (typically the lead compound in a project) to the ligand(s) of interest. As such, a "hub and spoke" map with the lead compound at the center of the graph and all other compounds directly connected to the lead compound might be a better approach for lead optimization in active drug discovery projects whereas a generic graph might be more useful for retrospective validation work where any given compound could be considered as the reference. Also note that in this case there are two separated graphs, which can occur when some ligands are too dissimilar to connect directly. In such cases, the experimental binding energy for at least one compound in each subgraph would be needed to compare the energies of the ligands in separate graphs.

dummy atoms are progressively perturbed to the actual atoms of the final molecule, fully appearing at its end state. Perturbations of this kind often necessitate more lambda intervals along the perturbation path to increase overlap between states and reduce convergence errors.

An important consideration and potential source for large errors arises from perturbations that require ring breaking, and often atom mapping protocols avoid ring breaking when possible.[163] It has been shown that errors in the bonded terms introduced from the dummy atoms are identical between the bound and unbound states and, therefore, should cancel out in the RBFE simulations. However, this criterion is only valid when the conformational ensemble of the molecule is not significantly affected by that of the remaining real atoms, which is not likely when the dummy atoms are coupled to the conformation of the remaining portion of the molecule, as is the case when they are members of a ring. Therefore, the free energies obtained from a restricted (and inaccurate) conformational ensemble will lead to the errors often observed in RBFE calculations involving ring making/breaking. Note that this is only relevant in the case where a perturbation creates an additional ring bond into the system, such as for macrocyclization or ring closing, but not for the case of adding a functional group with a ring, such as a phenyl, which is common in RBFE simulations. Recent work that includes a relatively simple "soft bond" potential suggests that improvements to core hopping scenarios can be made, although validation has been minimal.[164]

Size of the Perturbation. Obtaining accurate binding free energy predictions requires overlap in the phase space between neighboring lambdas along a perturbation path, and larger perturbations generally result in less overlap between the states.[67] While there are no hard rules about the size of a perturbation—it depends on the energy landscape of the protein–ligand complex, which is unique in every system—in practice, we find that perturbations of a few atoms can be handled routinely and larger perturbations may be possible, depending on the energy landscape of system. Conversely, in cases of very small changes, the ensemble explored by a single reference ligand may serve to calculate the free energy with respect to several other ligands, thereby expanding the chemical space that can be explored without additional computational costs.[165] This approach can be expanded by introducing a reference ligand (either real or fictitious) that is able to induce a conformational space in the receptor that is compatible with many ligands, thereby allowing free energy estimates to be computed via a single MD run rather than a separate simulation for each perturbation pair.[166,167] In general, the overlap between adjacent lambda windows cannot be predicted in advance of a simulation and therefore should be analyzed for each completed perturbation. When sufficient overlap is not attained, one can run longer simulations, use more lambda windows, or insert intermediate compounds into the perturbation map. More on this topic will be discussed in the section on analysis and convergence.

Covalently-Bound and Metal-Bound Ligands. In recent years, there has been a renewed interest in covalent inhibitors and metal-binding inhibitors.[168] Although molecular mechanics force fields cannot calculate the energetics of covalent bond formation, free energy simulations can still be used in such situations by analyzing only the noncovalent interactions, although the accuracy of such predictions depends on the fundamental assumption that the free energy change from the covalent bond formation is constant throughout the ligand series of interest. This assumption will be invalid in most cases where the covalent portion of the molecule is changing, and in some cases where delocalized effects from ligand modifications change the strength of the covalent interaction. Similarly, the relative affinity of metal-bound ligands can be directly compared if the contribution coming from the formation of

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the ligand–metal bond is close to constant among the ligands in the series and therefore cancels in the RBFE calculations, although such assumptions must be made with caution and be carefully validated. When the energetics of the covalent or metal interaction are suspected to vary across the ligands of interest, it should be possible to perform a separate quantum mechanical calculation to assess the energy of the covalent linkage, which can then be used to augment the RBFE calculations, although validation of this approach has not been published to our knowledge and is an interesting area for future research in the field.

Generating a Perturbation Map. The way all molecules in a congeneric series are connected in RBFE simulations can influence the results and therefore should be considered prior to running RBFE simulations. While the free energy equations show how to calculate the relative binding free energy between two different ligands, in practice one often wants to rank an entire congeneric series of compounds, which needs to be connected in some way so all energies can be compared (see Figure 6). A trivial solution to this task would be to perform perturbations to each ligand from a single reference compound. While straightforward, this may not be an optimal way to setup the perturbations, since RBFE simulations work best for transforming similar compounds, and a ligand series may not contain one reference compound that is highly similar to every other ligand. Choosing a sequential series of perturbations (e.g., arranging the compounds by similarity) is a possible way to connect molecules that minimizes the problem of non-overlapping lambdas, but suffers from error propagation, and therefore the accuracy of comparing two ligands becomes dependent on their location in the series of perturbations. In most cases, one should use the lead compound in the drug discovery series as the reference, since the ultimate goal of RBFE simulations will most likely be to improve the lead compound and not to minimize the errors across the entire graph, where the relationship between different idea molecules is less important than the energies of each molecule compared with the lead compound.

In general, it is useful to connect all ligands in a single graph in a way such that each ligand is connected to at least two other ligands with closed cycles to compute sampling error estimates (see Figure 6), as done for example in the LOMAP algorithm.[11] This allows for easy detection of sampling problems,[169,170] since the total free energy change in a closed thermodynamic cycle should sum to zero and deviations from zero reveal sampling errors along at least one leg of the cycle. Also of notable importance is that the redundant predictions from all legs of a closed cycle can be used to correct the free energy estimates and reduce the sampling error as compared with a single perturbation.[29]

Analysis and Convergence. Insufficient sampling is a potential source of error in all molecular dynamics studies. Improving simulation convergence by extending simulations to longer runtimes is a straightforward way to improve the accuracy of free energy predictions; however, a look at the relevant time scales and corresponding energy barriers[124] makes it clear that converging MD simulations of macromolecules by brute force is often impractical in the context of drug discovery projects, where results are often needed on a weekly basis to impact medicinal chemistry decisions. Due to the exponential dependence of the crossing rate between states and the height of an energy barrier in a simple Arrhenius formalism, extending a simulation by a factor of 10 would allow it to cross energetic barriers only ∼1.4 kcal/mol higher, at a cost of multiple weeks of wall clock time in the context of the calculations discussed here. As such, overcoming a barrier that is ∼5 kcal/mol higher than those being crossed during a certain simulation time would require more than 1000 longer simulations using standard MD. Despite rapid increases in available computing power and some encouraging reports of improving results in longer simulations,[171,172] extending MD runs is likely to be only a minor contribution to increased reliability of free energy calculations. Below, we discuss ways to analyze convergence and explore potential remedies in cases of unconverged simulations. When RBFE simulations are determined to be unconverged, it may be possible to apply enhanced sampling techniques to overcome the energy barriers. Many enhanced sampling methods exist, each of which tend to have advantages and disadvantages, and it is beyond the scope of this work to discuss the various enhanced sampling methods in detail. The authors refer the reader to several excellent reviews on enhanced sampling.[173−177]

Additional analyses should be performed on RBFE simulations to build confidence in predictions before making final recommendations for chemical synthesis. For example, traditional tools for MD analysis can be applied to look at atom movements and fluctuations of the ligand and protein. Cases with abnormally high values should be investigated for problems. In addition, the root mean squared deviation and fluctuations (RMSD and RMSF) for the protein (as a whole and around the binding site) should be investigated to ensure that the protein maintains its structure (more on this below). Large ligand perturbations can induce partial unfolding in proteins and lead to inaccurate results. RBFE simulations offer an additional challenge over standard MD simulations due to the multiple intermediate lambdas, but in general the two end points can be studied to gain insights into convergence. Specific care should be placed on studying the movements of the core of the molecules, which is assumed to maintain a common binding mode across the ligands in the series. If the initial and final lambda states have significantly different core positions then it is likely that the predictions will be inaccurate for that pair of ligands.

Estimating Sampling Errors. The Bennett acceptance ratio (BAR) method[43] is the most commonly used method to estimate the free energy difference from simulations, where the errors in the relative free energies can be calculated with the block bootstrapping method. In short, in block bootstrapping the data is divided into blocks of observations and is sampled randomly with replacement. Bootstrapping measures the computational uncertainty, which arises from the finite nature of the simulation, both in time and in space. A small bootstrap-estimated variance indicates good sampling within the local minima that have been explored. However, BAR cannot account for states that have not been observed, and thus running multiple independent calculations within a redundant perturbation map, as described above, or running multiple RBFE simulations for a given perturbation (e.g., with different random seeds and/or starting configurations) is a more robust way to estimate sampling errors (more on this below).

The major reason for the increased error with reducing simulation time is explained graphically by Chipot and Pohorille:[178] reduced sampling decreases the probability of conformational overlap between adjacent windows and this deteriorates the free energy estimate. Another valuable way to detect unconverged conformational sampling is by performing

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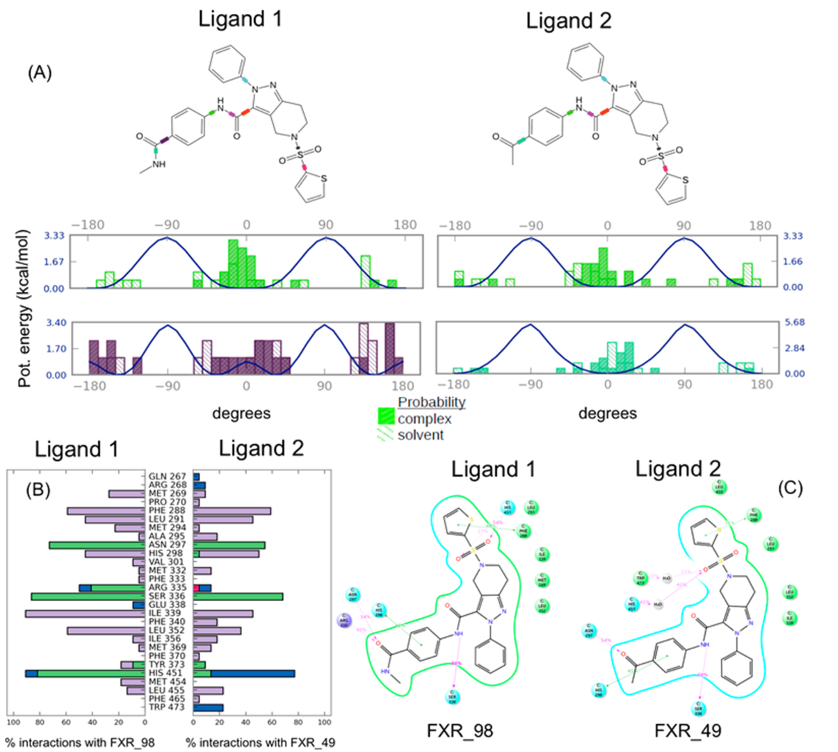

Figure 7. Example of a Simulation Interaction Diagram (SID) from an RBFE calculation. (A) Force field torsional energy profiles (blue curves) shown superimposed with the probability from the calculations in the complex (solid) and solvent (hashes) simulations for each of the two rotatable bonds in the ligands (green and magenta). (B) Comparison of interactions with different receptor residues between the two ligands. (C) SID for each of the ligands with residues contacting the ligands and the percent of the simulation spent making each interaction. Figure generated with the Maestro program.[181]

the same perturbation in reverse direction (i.e., preparing the system with ligand B and perturbing to ligand A rather than the reverse).[179] When a large discrepancy occurs in the free energy, this is a signal that a nonoverlapping ensemble has been sampled in the reverse simulation and that longer simulation times or enhanced sampling treatment might be required. Finally, sampling errors can be detected by starting independent calculations from different initial conformations that might be generated via sampling methods (e.g., docking, Monte Carlo, MD) to see if there is any starting structure bias. Alternatively, the same initial conformations with two different random seeds could be used, although differences between such simulations might not be as pronounced under short simulation conditions as compared with using different input conformations.

One way for checking the convergence of a free energy calculation is to plot the cumulative free energy differences, ΔG, over varying MD time ranges. The binding free energy should be converged sufficiently before the end of the simulation in order to have confidence that the important states observed within the simulation have been sufficiently sampled, although this approach would not detect cases of being stuck in local minima where alternate low-energy states have not yet been sampled. Another technique is to plot the free energy estimate for the solvent and complex legs individually as a function of simulated time, which could reveal the source of the unconverged results.[180] When the solvent leg is unconverged it is more tractable to run longer simulations only for the solvent leg, since the number of atoms is generally much smaller than for the complex leg and therefore simulations are much faster. In addition, applying enhanced sampling to the solvent leg is generally more straightforward, since the entire ligand can be included in the enhanced sampling region and details of the protein motions need not be considered. Plotting the reverse running average free energy calculated from the end of the simulation toward the beginning can be also helpful in highlighting changes that occur late in the simulation, where a forward average minimizes late changes due to the average over many previous states. Observing a significant change toward the end of the simulation can sometimes be addressed with only slightly longer simulations, since the transition has already been observed at least once and therefore is energetically accessible within the time scale of the simulation. Finally, as mentioned before, perturbation maps can be designed with closed cycles, which can highlight incongruences due to limited sampling,[169] although this does not point to the single problematic perturbation within a cycle. In this respect, error models using multiple cycles can be useful in identifying the precise location of erroneous perturbations.[29]

Trajectory Analysis and Visualization. In addition to the final computed free energy values from RBFE calculations, the trajectories offer a wealth of information that can be analyzed and visualized to extract additional insights. Statistical analysis of trajectories can facilitate a quick assessment of the quality of a simulation (RMSD over time, per-residue RMSF, etc.). Such information can be plotted to identify, for example, parts of the protein or ligand that are unstable or highly mobile. Of particular interest are parts of the system that are different between the initial and final states in the RBFE simulations. Often such plots alone are insufficient to gain deep insights, and visualization of the molecular structures evolving over the trajectory can be useful. In some cases, high conformational

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mobility is a real property of the system, but sometimes it can also be an artifact arising from poor system setup, inadequate simulation equilibration, force field inaccuracies, or errors in the system. In the cases where the target flexibility is real, this information can be used in the discovery process, for example, to identify transient pockets that could be targeted with ligand modifications. Simulation analysis can be challenging due to the many lambda windows along the perturbation path. In most cases, it is sufficient to analyze only the first and last lambda values, which correspond to the two end points and represent physically meaningful systems (one containing the initial and one the final molecule) in order to see differences in the simulations.

Beyond trajectory visualization, additional information can be extracted from the trajectories. For example, the Simulation Interaction Diagram (SID) module of the Maestro program generates a report of information related to a trajectory (see Figure 7). Figure 7a shows a breakdown of ligand torsions, which could highlight sampling problems or reveal ligand strain induced by binding to the protein. Relieving such torsional strain could lead to a boost in potency, thus providing an idea to pursue for rational design. A useful way to compare between multiple trajectories is by looking at differences in protein–ligand interactions over the simulation. While a thorough analysis would involve pairwise (and higher order) correlations, a simple analysis of the frequency at which different interactions are made and how those differ between the initial and final molecules can be revealing (see Figure 7b). Similarly, those interaction frequencies can be mapped back to a structural representation, adding additional insights about the ligand atoms involved in the interactions (see Figure 7c).

While many other visualization ideas have been presented in the literature, it should be noted that there is still significant room for advances in this area. For example, quickly detecting dissimilarities in dynamics between different simulations can be challenging, even for experts in the field. Additionally, extracting energetic contributions to the total free energy would be extremely valuable but cannot be done rigorously, since many of the energy terms are not pairwise decomposable (e.g., entropy and solvation). Nonetheless, even without a rigorous decomposition, an estimate of the thermodynamic contributions of different parts of the system would be desirable (e.g., how much energy is coming from a particular ligand hydroxyl interacting with a side chain of interest or how much strain is in ligand A versus ligand B). The more information beyond a binding energy that can be extracted from the calculations, the greater value they will have in drug design.

Comparing Computed Results with Experiments: K_i vs IC_{50}. To directly compare the relative potencies between inhibitors, a quantitative measure must be employed. The $\Delta\Delta G^0$ values predicted by relative binding free energy calculations are directly comparable to K_i values, but often only experimental IC_{50} values are available. While IC_{50} values and K_i values are sometimes reported interchangeably in the literature, direct comparison of these values is not always possible. The relationship of K_i and IC_{50} for a given compound varies depending on the assay conditions and the compound's mechanism of inhibition. The Cheng–Prusoff equations describe this relationship mathematically for competitive inhibitors that bind to the free enzyme:[182]

$$\mathrm{IC}_{50} = K_i\left(1 + \frac{[S]}{K_\mathrm{m}}\right) \tag{3}$$

For noncompetitive inhibitors that bind only to the enzyme–substrate complex, the relationship becomes

$$\mathrm{IC}_{50} = K_i\left(1 + \frac{K_\mathrm{m}}{[S]}\right) \tag{4}$$

Matters become more complicated when there is an intermediate behavior between purely competitive inhibition and purely noncompetitive inhibition; these inhibitors bind to both free enzyme and enzyme–substrate complex, with inhibition constants K_ie and K_ies, respectively.[183]

$$\mathrm{IC}_{50} = \frac{[S] + K_\mathrm{m}}{\left(\frac{[S]}{K_\mathrm{ies}} + \frac{K_\mathrm{m}}{K_\mathrm{ie}}\right)} \tag{5}$$

Relating IC_{50} values to K_i without taking into account the above parameters can be challenging and misleading as these two quantities have been shown to differ by as much as 400-fold.[184] It should be stressed that this comparison is only possible if the mechanism of inhibition and the substrate concentration are known. Then, IC_{50} values can be converted into K_i values using the Cheng–Prusoff equations. If the concentration of substrate used in the assay is not provided, in the case of competitive inhibition, one may only conclude that IC_{50} values approximate K_i when the $[S]$ used in the assay is much lower than K_m. Strictly speaking, the IC_{50} is always higher than K_i, thus any reported IC_{50} value is an upper limit for the K_i of the compound. For noncompetitive inhibitors, IC_{50} values approximate K_i when the $[S]$ used in the assay is much higher than K_m. At large values of $[S]$, the ratio $K_m/[S]$ approaches zero, and IC_{50} approaches K_i (eq 4).[183] Different methods have been proposed to predict the IC_{50}, and its standard error, at any chosen substrate concentration, thereby facilitating comparison with results obtained with similar inhibitors, homologous enzymes, and interlaboratory comparisons.[185] Fortunately, for relative energy comparisons (as in RBFE calculations) the $[S]/K_m$ and $K_m/[S]$ terms in eqs 3 and 4, respectively, cancel between two ligands and therefore relative energies extracted from IC_{50} values can be directly compared, if $[S]$ in the experiments is the same for the different ligands.

To complicate matters further, covalent or slow-binding inhibitors are incubation time-dependent and their K_i and IC_{50} values are not comparable between different assay protocols. Ignoring these issues could lead to wasted effort with compound optimization and difficulties in developing accurate scoring functions on such data.[184] As mentioned in the work of Cisneros et al.,[184] standardized control screenings should be performed using compounds with aqueous solubilities over 10 μM and hits need to be tested for time-dependent inhibition. When possible, K_i values should be sought and IC_{50}-values avoided, reducing the ambiguities in experimental results and allowing for comparison of results from different laboratories. Finally, it should be noted that the dynamic range of binding free energies in experimental measurements is often only a few kilocalories per mole in lead optimization for a given round of modifications. Therefore, when comparing computed binding free energies and experimental data it may be difficult to obtain low RMS errors compared to experiment even with a computational method that predicts accurate binding affinities.[186]

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As already mentioned, RBFE calculations are frequently compared to IC_{50} values obtained mostly from biochemical assays, despite the inherent limitations and approximations in this comparison (see above). As an alternative, when K_i are not available, several biophysical methods commonly used in the pharmaceutical industry can also provide affinity measurements to assess the RBFE predictions. In particular, isothermal titration calorimetry (ITC) would be the experimental method of choice for a direct comparison of $\Delta\Delta G^0$ values predicted by RBFE simulations. This biophysical method provides K_d values in the range of 1 nM to 100 μM, is label-free and does not require any immobilization. However, the large amount of protein required for each K_d measurement, the necessity to have a protein highly soluble, and the throughput restricted to 10s ligands per day highly limit its usage.[187] Surface plasmon resonance (SPR) is another biophysical method that provides K_d measurements for protein−ligand binding with high sensitivity and with higher throughput (100s ligands per day) and much lower protein consumption compared to ITC. However, this technique requires the immobilization of the protein on the sensor chip, which might interfere with binding. It is therefore important that the binding of the ligand to the protein should not be affected by this immobilization.[187]

Limitations. While free energy methods offer great promise, there are still a number of challenges, many of which have been discussed above and will be summarized here. For example, although free energy simulations have been shown to work on homology models,[72] in general the input structure has to be of sufficiently high quality such that the per-lambda MD sampling can reach the relevant local minima. This typically means having a crystal structure of sufficiently high resolution to accurately place all binding site residues and a cocrystallized ligand from the ligand series of interest. When this is not available, calculations could still provide predictive results, but extra care must be taken to generate good protein−ligand complexes as starting structures for free energy simulations. Even with a high-quality initial model, convergence in the free energy calculations will depend on the free energy surface of the system of interest. In general, large-scale protein movements (e.g., DFG-in/out in a kinase) cannot be sampled sufficiently within the time frame of standard free energy calculations (1−10 ns per lambda), but could be sampled with longer simulations or enhanced sampling techniques. As previously shown via an Arrhenius-like argument, even small-scale conformational changes can be inaccessible if the barrier is too high. For example, an amide bond has a high torsional rotation barrier that would make it unlikely to flip within a given lambda window. If such specific degrees of freedom are thought to be important, then enhanced sampling methods should be applied.

In addition to sampling, several other factors pose challenges to free energy methods. Perturbations that introduce a change in formal charge, as discussed above, are known to be difficult. Another potential challenge, which is related to sampling, is the treatment of explicit water molecules. In many cases, buried waters cannot exchange freely with bulk solvent over the time course of a short MD simulation (∼5 ns), but introducing ligand modifications into these regions of buried waters require simultaneous changes in the number of buried water molecules in order to accurately predict binding energies. Finally, even with complete sampling, performance can be limited due to force field errors/inaccuracies, which include not only parameters but also fundamental limitations built into the molecular mechanics force field equations. In many cases, inaccurate ligand parameters can be resolved through adding new, more specific parameters (generally based on quantum mechanical torsion scans). In other cases, the functional form of the force field would need to be reconsidered to get a better representation of the approximated quantum mechanical nature of the atoms and bonds. One such approach is to add off-centered charges (virtual sites) that can be used to represent lone pairs or other nonspherical charge distributions (such as the sigma hole in halogen bonds).[116,117,188]

■ EXAMPLES AND APPLICATIONS

Using free energy simulations for molecular design dates back to the mid-1980s when the first free energy calculations were performed to calculate the difference in free energies of hydration between two molecules[38] and soon thereafter it was first applied to protein−ligand binding.[189] Since then and until very recently, applications were typically limited to a handful of ligands that were analyzed after the experimental work (retrospective validation) because the calculations were computationally too intensive to achieve convergence with the available computational resources to be used in prospective applications. While retrospective studies often report good results, the quantity of data is insufficient to draw statistically significant conclusions and there are often questions about the applicability of retrospective studies to prospective work. Fortunately, in recent years more prospective free energy simulations have been published, with the total number of ligands reported either in peer-reviewed publications or scientific presentations now in the thousands, covering dozens of targets. In general, the results are consistent across these studies, with errors around 1.0 kcal/mol and good correlation with experiments in favorable situations. Nonetheless, considerable data also exists showing that free energy simulations still yield unsatisfactory results in many cases, but these results are often confined to presentations and not published, and therefore are not covered in this review. Needless to say, there is still room for improvement in many aspects of free energy simulations, including force fields, sampling, setup, and analysis. While we cannot cover all free energy simulation studies in this review, we attempt to summarize a variety of examples, which combine retrospective and prospective applications across a broad range of target classes and ligand perturbations using a variety of free energy methods with the aim to provide a summary of the current state of the field and how to maximize chances of success on a given project.

Some of the first studies that demonstrated the potential of free energy simulations to be used in small molecule optimization were performed by the Jorgensen group, where the first protein−ligand binding calculations for lead optimization using the RBFE approach appear in the literature, in this case using the program MCPRO for calculating the free energy difference between cyclosporin A variants binding to human cyclophilin and benzamidine inhibitors binding to trypsin.[190,191] Subsequent works by the Jorgensen group were carried out on the affinity/selectivity of celecoxib variants binding to COX-2/COX-1,[192,193] heterocycle optimization of ligands binding to fatty acid amide hydrolase (FAAH),[194] and the effects of HIV-RT mutations on the activities of non-nucleoside reverse transcriptase inhibitors (NNRTIs).[195−198] More recent examples using MCPRO investigated the methyl effects on protein−ligand binding for methyl replacements in inhibitor series for p38α MAP kinase, ACK1, PTP1B, and

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thrombin, where it was shown that addition of a methyl group can provide an activity boost up to a 100-fold increase in potency, owing to coupling the conformational gain with the burial of the methyl group in a hydrophobic region of the protein.[199] These studies, while primarily retrospective calculations on molecules with known binding affinities, were instrumental in laying the foundation for the field and establishing the potential value that free energy simulations can provide in the hit-to-lead and lead optimization stages of drug discovery.

In a recent study performed at Boehringer Ingelheim, free energy calculations on a test set of ∼100 compounds binding to five targets (phosphodiesterase 5 and four undisclosed in-house systems) were analyzed.[22] The TI formalism in the AMBER suite[200] along with an automated setup procedure creating a maximum spanning tree of ligand connections was employed. This study found no major differences in results regarding the exact free energy formalism used, but instead, a strong system dependence of the calculation accuracy was observed, with per-system errors ranging from 0.5 to 2.0 kcal/mol. As an estimator of calculation uncertainty, the standard deviation from independent runs was used. Interestingly, the study found no significant correlation between the size of the perturbation performed and the resulting uncertainty estimate or the deviation from experiment. This suggests that the relationship between accuracy and the perturbations is not simple. Indeed, beyond the size of the perturbation, it will likely be important to analyze other properties of the perturbation (shape, electrostatics, location of the perturbation in the binding site, protein flexibility around the perturbation, charge induction effects, etc.) in order to better assess factors related to the performance/accuracy of RBFE predictions. A small perturbation into a part of the pocket with a high degree of flexibility may take longer to converge than a large perturbation into solvent. Additionally, ligands with incorrect initial poses, especially those that make incorrect hydrogen bonds with the protein or have incorrect solvation in buried pockets, may lead to increased convergence time due to being stuck in the wrong local minima. In general, convergence will depend on the characteristic time scales for the relevant motions involved in the binding process within the system of interest. Understanding the time scales for important motions in protein−ligand binding is a current challenge in the field where advances would provide great value.

Essex and co-workers also provided validation of free energy simulations for multiple targets. In an earlier study on trypsin they reported a mean unsigned error (MUE) below 1 kcal/mol,[191] while more recently they studied COX2, neuraminidase, and CDK2, yielding MUEs of 0.76, 3.31, and 4.55 kcal/mol, respectively.[201] The authors ascribed the low accuracy in the case of CDK2 to force field limitations and pose ambiguity, although conformational sampling and system setup could have been an issue as well. For example, the calculations were performed on CDK2 without cyclin, while as the name implies, CDK2 (cyclin-dependent kinase 2) depends on cyclin, so running without cyclin could present problems. Such system setup issues can be more important in free energy simulations compared with more traditional SBDD techniques like docking, where atoms far from the binding site typically do not contribute to the results. In another study by the Essex group on the estrogen receptor β ligand binding domain they were able to successfully discriminate hits from decoys.[202] Additionally, they benchmarked extensively implicit solvent models compared with explicit solvent and determined that they could gain considerable speedup and similar results using implicit solvent models.[201] They also found free energy simulations are always more predictive than a standard scoring function. These studies were among the first to show that approaches could deliver results in a time frame sufficient to be used in drug discovery projects. However, one should not draw broad conclusions based on limited data about important aspects of the calculations, such as the solvation model. The difference between implicit and explicit solvent is considerable from a theoretical perspective and there are certainly cases where implicit solvent models will not perform well, such as systems where there are buried isolated water molecules that do not behave like bulk solvent. While early studies suggested that implicit solvent models might be sufficient,[203] more recent evidence from peptide conformational studies and protein folding simulations suggest that implicit solvent models are insufficient to yield correct free energy landscapes.[204−206] Perhaps such cases could be handled with a mixed approach, where certain important water molecules are handled explicitly whereas the majority of the system can be treated with an implicit solvent model, yielding the speed of an implicit model but retaining the important characteristics of explicit waters, although more studies are needed to derive any broad conclusions along these lines.

In another class of pharmaceutically relevant targets, free energy simulations have played an important role in elucidating the ion specificity in channels and transporters.[207] By perturbing K^+ into Na^+ and analyzing the coordinating carboxylic ligands, Noskov and Roux were able to propose a mechanism that explained the KcsA potassium channel selectivity for K^+ with respect to Na^+ and showed the complex electrostatic interdependence between ligands and ions in this channel. Similar studies on the NaK channel were used to explain the lack of specificity for Na^+ and K^+.[208] Interestingly, a similar strategy showed the different origin of selectivity in the case of two Na^+ binding sites for LeuT transporter[209] and explained the Na^+ and H^+ selectivity for different ATP-synthases in various organisms.[210,211]

Roux and co-workers also investigated the specificity of Gleevec toward Abl Kinase compared with c-Src,[212] where they combined results from absolute binding free energy calculations with those from umbrella sampling calculations for estimating the free energy of the DFG-flip and found that the specificity is driven by a combination of conformational selection and binding stabilization. The binding affinity was determined to arise from a stabilizing contribution of vdW in the case of Abl. The same strategy was applied to explore selectivity differences of Gleevec binding to c-Kit/Lck kinases, and the lack of selectivity of the G6G ligand binding to Abl/c-Src kinases.[212] While this work is encouraging, it is by no means routine to compute the free energy difference between protein conformations and highlights another area where additional work, both in terms of the scale of the validation and methods to make such calculations more routine, would be greatly valued.

In a nonpeer-reviewed publication, Bayer Pharma AG reported the use of free energy calculations in drug discovery projects, running hundreds of compounds across more than ten different targets (a mix of retrospective and prospective cases).[23] Their study concluded that modern free energy calculations have achieved the speed and usability necessary to be useful in driving projects, although accuracy and consistency could still be improved. For a retrospective analysis of 147

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calculations, mean unsigned error of 0.94 kcal/mol and an R-value of 0.77 were reported. Among the remaining challenges identified were finding accurate force field parameters for novel chemical groups, conducting complex scaffold transformations, and insufficient water sampling. Interestingly, the authors highlighted examples of compounds that were synthesized only because of the free energy predictions, which ended up advancing projects, demonstrating significant value. While the prospective application of free energy calculations in drug discovery projects is still limited, we expect to see a growing number of studies like this, where novel molecules are prioritized for chemical synthesis based on the results of free energy calculations.

A number of retrospective and prospective applications of free energy simulations in drug discovery projects have been published by Janssen Pharmaceuticals,[25,171,213] where the researchers used the FEP+ software package from Schrödinger on an in-house GPU cluster. For PDE2, RBFE calculations were used to probe ligand SAR in an effort to develop a potent and selective pyridotriazolopyrazine compound with low nanomolar affinity against the human PDE2 but not the related PDE10 enzyme. In addition, running standard MD simulations to probe differences in protein flexibility provided a better understanding of the ligand-induced opening and closing of a small binding pocket via protein side chain conformational changes (see the SI of ref 25), which illustrates that RBFE calculations alone are not a panacea and are generally most valuable when coupled with other computational and experimental approaches.

The study by Janssen on β-secretase (BACE) combined both retrospective validation of R-groups targeting a specific subpocket of the BACE1 active site and prospective triaging of novel compounds with similar modifications for synthesis. Furthermore, the effect of longer MD simulations was explored, showing average errors decreasing from 0.9 to 0.6 kcal/mol when simulations times were increased (from 5 to 20 ns/λ). In general, longer simulations produce better results for RBFE calculations, as seen here, but the optimal simulation time depends on a number of factors, including the characteristic time scales of the relevant motions in the system as well as the throughput required to impact synthesis priorities. A later report by the same group demonstrated the applicability of FEP+ calculations to predict the effect of changes in scaffold ring size for binding to another part of the BACE1 binding pocket.[213] Changes in molecular topology, such as increasing flexible ring sizes, have historically been considered challenging for free energy methods and benefit from recent algorithmic improvements.[164] The BACE1 inhibitor design study combined large scale free energy simulations with an exploration of additional sampling effects and retrospective validation on known compounds closely resembling the prospective molecules to be studied. This approach is a good example of how simulation tools can be used in an applied drug design context, where target and binding mode uncertainties are common and project time lines rarely allow iterative refinement of results.

In addition to the earlier retrospective work from the Jorgensen lab presented above, more recent studies from the same group have demonstrated the value of free energy calculations using MCPRO in the prospective discovery of anti-HIV compounds.[125] The first reports of rapid lead optimization non-nucleoside inhibitors of HIV-1 reverse transcriptase (HIV-RT) from micromolar leads to the nanomolar level appeared in 2006,[214−216] followed by an interesting example where a false positive from a virtual screen could be optimized into a true active non-nucleoside reverse transcriptase inhibitor (NNRTI) against HIV-RT using MCPRO.[217] Subsequent efforts in recent years resulted in novel NNRTIs with reported picomolar and low-nanomolar activities against wild-type HIV-1 and key variants,[16,218−220] which also showed improved solubility and lower cytotoxicity than recently approved drugs in the class. Numerous drug design issues were successfully addressed by coupling the simulations with synthesis, biological assays, and protein crystallography to improve potency and solubility against a wide range of viral variants. In fact, the most potent HIV-RT inhibitor reported to date has arisen from this work.[125]

Lovering et al. at Pfizer have also employed the MCPRO approach to optimize compounds, this time to spleen tyrosine kinase (Syk).[24] Specifically, they optimized an imidazopyrazine core using a combination of computational approaches, including free energy simulations. First, a small set of compounds was used to retrospectively validate the MCPRO approach on this target, which resulted in 12 of 13 RBFE predictions having the correct sign. MCPRO was then applied prospectively to evaluate and prioritize new cores (17 in total), one of which yielded a 10-fold improvement in activity relative to the parent compound. Further optimization of the new core led to compounds with nanomolar cellular activity. This example shows the prospective use of free energy simulations both for core hopping and R-group optimization.

As a final example, researchers at Schrödinger showed that the throughput of RBFE simulations was generally fast enough to impact active drug discovery projects, as demonstrated in the case of their JAK/Tyk2 inhibitor program, where FEP+ was utilized throughout the project to prioritize chemistry deicisions.[221] Previous work by the same group and academic collaborators showed that it is possible to consistently produce errors of less than 1 kcal/mol across a broad range of protein targets and ligand perturbations, with eight retrospective series and two prospective projects.[27]

■ CONCLUSIONS

The promise of using physics-based molecular simulations to improve the efficiency of the preclinical drug discovery process has generated renewed interest in recent years, fueled by decades of scientific advances coupled with vast increases in computational power. Indeed, the ability to accurately predict the binding affinity, *in silico*, of a ligand to a target (or off-target) protein would be highly valuable in structurally enabled lead optimization efforts. However, there are many scientific and technical challenges that must be considered in order to reliably obtain accurate results in a timely enough fashion to impact projects. In this perspective, we presented the state of relative binding free energy (RBFE) simulations with a focus on recent drug discovery applications (retrospective and prospective), highlighting key considerations for determining whether RBFE calculations could add value on a given project. We summarized the underlying theory behind RBFE simulations along with recommendations for best practices for applications and remaining challenges for the field to ponder.

RBFE simulations present a particularly timely topic for a review due to the recent rise in visibility resulting from high-profile publications and stories of real-world successes. To illustrate this upsurge in popularity, a simple Google Scholar search for “free energy perturbation” AND “binding” at the beginning of 2017 revealed only a single hit in 1980, 65 in 1990, 324 in 2000, 773 in 2010, 844 in 2015, and 918 in 2016.

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While increasing citations are an indicator of general interest, it is the real-world applications that will ultimately dictate the impact of this approach. Thus far, results for prospective applications of free energy simulations are encouraging, as illustrated by some of the cases presented here, but still limited. However, it is still too early to know how valuable these methods will prove to be in drug discovery.

While most RBFE publications present favorable results, there are a number of limitations and challenges that are typically not presented in publications and must be considered to fully grasp the current (and future) value of RBFE simulations in drug discovery. First, it is essential to start with a reasonably good structural model of a protein−ligand complex, where the bound ligand is similar to the series of interest. In most published examples, the methods have been applied to crystal structures where a holo structure exists with a ligand from the congeneric series of interest, in which case one can be reasonably confident in the binding mode of related compounds in the series. However, even with a cocrystal structure, success is not guaranteed with RBFE simulations. Many confounding factors can lead to poor performance, which generally fall under the broad categories of incorrect system setup, poor force field treatment, and insufficient sampling, the details of which have been discussed in this work.

When a cocrystal structure is not available, as is common in drug discovery projects when RBFE calculations could be of the most use, then docking or some other technique must be applied to generate an accurate starting structure for the simulations. Predicting the binding mode of a novel ligand to a target can be a formidable challenge, as demonstrated by many docking comparison studies and blind challenges.[222,223] In real-world applications, one would use any available data to validate a putative docking pose (distance constraints, known SAR, mutagenesis studies, etc.), so perhaps the accuracy of cross docking in prospective projects is higher than the published validation results, where the use of experimental constraints is typically avoided in order to directly assess the method, independent of human-introduced constraints. In addition, there could be opportunities to improve docking programs in a systematic way when the primary objective is pose prediction as a starting point for RBFE simulations as opposed to the more traditional role of docking in virtual screening or pose prediction of diverse molecules. Fortunately, the sampling in RBFE calculations can overcome imprecise docking calculations, so long as the correct pose can be accessed within the time scales of the simulations. In general, the closer one starts to the correct binding mode the more accurate will be the results and the faster convergence will be achieved.

Regarding sampling, in addition to the degrees of freedom sampled by molecular dynamics (i.e., the evolution of the atomic coordinates of all atoms in the system over time), one must also consider degrees of freedom that are not sampled within traditional MD packages, such as protonation/tautomeric states, buried waters, and the location of counterions, to name a few. Some of these factors can be addressed directly with additional methods that combine with MD, such as using Grand Canonical Monte Carlo (GCMC)[169,224] to place buried water molecules or ions that cannot diffuse within the time scale of the simulations or constant pH MD[225] to sample tautomeric states. However, most RBFE implementations assume that these degrees of freedom are determined in advance of the RBFE simulations. There are a number of software tools that attempt to automate various aspects of the system setup, although it is recommended that great care be taken in the system setup, including careful manual review, in order to avoid costly calculations being performed on an incorrect system.

Although in this article we have focused on RBFE calculations, it is worth noting some other applications of free energy simulations that could be useful in the context of designing new drugs. First, absolute binding free energies (ABFEs) would allow for a direct comparison of binding between structurally unrelated molecules, thereby facilitating one of the greatest challenges in early discovery (e.g., scoring structurally diverse molecules binding to a target of interest and/or comparing binding energies to different targets). However, computing absolute energies generally takes much more computational resources than relative calculations due to the much larger phase space that must be sampled, and less cancelation of errors. We expect to see an increase in ABFE calculations in drug discovery as computational power continues to increase and further methodological advances make these calculations more efficient.

In addition to computing binding free energies, there are many other potential applications of free energy simulations—whenever one can define the two end points of a system it should be possible to compute relative free energy differences. As such, it opens the door to a broad range of other applications, such as solubility (solid versus aqueous phase),[226] protein−protein interactions (bound versus unbound),[26,172] permeability (aqueous phase versus embedded in the membrane),[227] and more. We feel that the application of free energy simulation approaches will continue to expand as we see the cost of computational resources decrease and the price of developing drugs rise.

Regarding RBFE simulations, there are several advances that would add considerable value in drug discovery applications. For example, charge-changing perturbations are notoriously challenging and generally produce errors that are too large to be useful in drug discovery. While handling charge changes would be helpful in small molecule projects, perhaps the greatest value would come in biologics projects where charge-changing modifications are more common when mutating among amino acids. Along these lines, advances in free energy calculations with alchemical perturbations applied to proteins would open new opportunities in the rapidly growing biologics space, and would facilitate predicting the energetic consequence of protein mutations on ligand binding, which has implications in drug resistance and personalized medicine. While free energy simulations have already been shown to work in some cases on protein residue modifications,[26,228] such validation is still limited and the published examples tend to have larger errors than small molecule modifications, indicating room for improvement. Another class of problems that needs further validation work involves calculations on macrocycles and cyclic peptides, which have received great interest in the biotechnology industry in recent years but have not been the subject of many free energy studies. Perhaps one of the greatest challenges for studying large ring structures will be the sampling of the unbound state of these highly constrained systems, which typically have high energetic barriers between different conformational states, whereas for typical small molecules sampling of the unbound state is relatively straightforward to converge.

Once accurate binding affinity predictions to a target of interest can be achieved, it is possible to move beyond affinity

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to explore selectivity (differential affinity toward/against multiple targets). Selectivity screening could be used to scan for liabilities that a molecule might contain when interacting with related subtypes to the target of interest or other ADME/Tox liabilities related to ligand-protein binding. Predicting selectivity could be pursued in either of two ways: (1) In the case where the targets are very similar (subtypes of a target or resistance mutations), it is possible to perform alchemical perturbations to the protein to directly assess the differences in binding of a compound to different receptor forms. (2) In the case where there are large differences between the proteins, directly perturbing the system will be unlikely to succeed, so the ligand series can be explored in parallel across different targets and then energy between the targets can be calibrated using experimental data if just one compound in the series has experimentally determined binding affinity data to all of the proteins of interest. Computing multitarget binding free energies could also be used to explore polypharmacology (i.e., intentionally binding to more than one target).

In summary, we believe that we are entering a new era for relative binding free energy calculations in which applications will transition from retrospective studies on a small number of compounds primarily in academic settings to large-scale prospective work that will drive drug discovery projects. Not all projects will be amenable to free energy simulation techniques, but the domain of applicability will continue to grow as structural biology efforts (both with X-ray crystallography and other methods, like Cryo-EM) solve more structures and the computational approaches improve (scientifically, technically, and practically). The examples highlighted in this review, although by no means a complete survey of the field, show the great promise of free energy methods. We expect to see a growing number of disclosed successes, although many will not be published until years after the success has been demonstrated in the project, as is typically the case in pharmaceutical research. Nonetheless, we are beginning to see the upturn in prospective applications showing that *in silico* RBFE predictions can be performed with the throughput and accuracy needed to improve the efficiency of drug discovery efforts by opening a much larger chemical space to computational explorations, which then get prioritized for synthesis, and we expect this trend to continue.

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